

EDITABLE TEST

Algebra 2
**MID-YEAR
EXAM
& STUDY GUIDE**

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Algebra 2 Mid-Year Test

Study Guide

Name: _____

Date: _____ Block: _____

Topic 1: Radicals

Simplify each expression.

1. $\sqrt{243a^{25}}$

2. $\sqrt[3]{-18p^9} \cdot \sqrt[3]{4p^5}$

3. $\frac{\sqrt[4]{32a^{13}b^5}}{\sqrt[4]{2a^2b}}$

4. $3\sqrt{72} - 2\sqrt{8} + 2\sqrt{20}$

5. $\sqrt{2}(\sqrt{2} - 6\sqrt{10})$

6. $\frac{6}{\sqrt{12}}$

7. $\frac{\sqrt{2} - 2\sqrt{3}}{\sqrt{8}}$

8. $\frac{2 + \sqrt{3}}{5 - \sqrt{3}}$

Topic 2: Rational Exponents

Write in radical form. Simplify if needed.

9. $150^{\frac{1}{2}}$

10. $(-32)^{\frac{1}{3}}$

11. $x^{\frac{11}{4}}y^{\frac{5}{4}}$

Write in exponential form.

12. $\sqrt[3]{64}$

13. $\sqrt{(xy)^5}$

14. $\sqrt[4]{10r^7s^{12}}$

Simplify. Write all answers in simplest radical form.		
15. $n^{\frac{7}{2}} \cdot n$	16. $\frac{k^{\frac{15}{4}}}{k^{\frac{1}{2}}}$	17. $\left(8^{\frac{4}{3}}\right)^{\frac{1}{2}}$

Topic 3: Complex Numbers			
*Recall: $i =$ $i^2 =$ $i^3 =$ $i^4 =$			
Simplify each expression.			
18. $\sqrt{-81}$	19. $2\sqrt{-14} \cdot \sqrt{8}$	20. $18i^{49}$	21. $(i^3\sqrt{5})^2 \cdot 2i$
22. $(4 - 7i) - (5 + 2i)$	23. $(9 + 2i)(4 - i)$	24. $(6 - 2i)^2$	
25. $\frac{15}{6i}$	26. $\frac{1 - 5i}{2i}$	27. $\frac{3 - 3i}{7 - i}$	

Topic 4: Classifying Numbers & Identifying Properties			
Simplify (if possible), then name all sets to which each value belongs.			
28. $\frac{\sqrt{64}}{5}$	29. $\sqrt{-36}$	30. $\sqrt{3} - \sqrt{3}$	31. $ 1 - 4^2 $
32. $2 + 5i$	33. $\sqrt{15}$	34. $-\sqrt{\frac{45}{5}}$	35. $-8i \cdot 5i$

Name the property that justifies each statement.	
36. $\sqrt{5}(8 - \sqrt{3}) = 8\sqrt{5} - \sqrt{3} \cdot \sqrt{5}$	37. $2m^3 + (-2m^3) = 0$
38. $-9 \cdot (2 + 3i) = (2 + 3i) \cdot -9$	39. $\left(\frac{x}{5}\right) \cdot \left(\frac{5}{x}\right) = 1$

Topic 5: Factoring		
Identify the special factoring patterns.		
DIFFERENCE OF SQUARES $a^2 - b^2 =$	SUM OF CUBES $a^3 + b^3 =$	DIFFERENCE OF CUBES $a^3 - b^3 =$
Factor the expressions below completely.		
40. $36x^2 - 64y^2$	41. $m^3 + 64$	
42. $81k^4 - 3k$	43. $2xy^5 + 250xy^2$	
44. $p^3 - 10p^2 + 25p$	45. $x^4 - 11x^2 + 28$	
46. $4x^3 + 18x^2 - 10x$	47. $9x^3 - 63x^2 - x + 7$	

Topic 6: Dividing Polynomials**Find the quotient by long or synthetic division.**

48. $(a^3 + 6a^2 + 2a - 20) \div (a + 2)$

49. $(x^4 - 64x^2 - 7x + 61) \div (x - 8)$

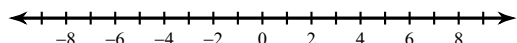
Topic 7: Absolute Value Equations & Inequalities**Solve each equation. Check all solutions.**

50. $|4m + 5| = 9$

51. $-5|7 - x| + 6 = -14$

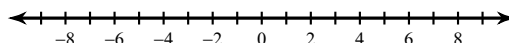
Solve each inequality and graph your solution. Write the solution in interval notation.

52. $|x - 2| \leq 3$



Interval Notation:

53. $-2|2x - 1| < -14$



Interval Notation:

Topic 8: Quadratic Equations

Discriminant Formula:

- If $d < 0$, there are **2 imaginary roots**.
- If $d = 0$, there is **1 rational root**
- If $d > 0$ and a **perfect square**, there are **2 rational roots**
- If $d > 0$ and **not a perfect square**, there are **2 irrational roots**.

Find the discriminant of each equation. Then, determine the number and type of roots.

54. $x^2 - 7x + 15 = 0$

55. $2x^2 - 72 = 0$

56. $-x^2 + 9x + 23 = 0$

Solve using the most appropriate method. Simplify all irrational and complex solutions.

57. $2x^2 - 18 = 78$

58. $x^2 - 16x + 73 = 0$

59. $9x^2 - 18x = 11$

60. $2x^2 - 10x - 26 = x - 5$

61. The roots of a quadratic equation are -7 and 1. Write an equation that could represent this function. Give your answer in standard form and factored form.

Topic 9: Polynomial Equations

Solve each equation. Simplify all irrational and complex solutions.

62. $x^3 - 27 = 0$

63. $2x^3 - 5x^2 + 18x - 45 = 0$

- 64.** A polynomial function is defined by the equation below. Identify the zeros, their multiplicity, and the effect on the graph

$$f(x) = -2x(3x + 2)^7(x - 1)^2$$

Zero	Multiplicity	Effect

- 65.** The zeros of a polynomial equation are -4 (multiplicity 2) and 3 (multiplicity 1). Write an equation in standard form that could represent the function.

Topic 10: Radical Equations

Solve each equation. Check all solutions.

66. $\sqrt{8x - 7} + 10 = 15$

67. $\sqrt[3]{2x + 12} = 4$

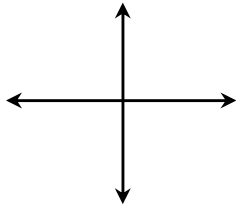
68. $\sqrt{-7 - 4m} = \sqrt{2m + 11}$

69. $\sqrt{56 - a} = a$

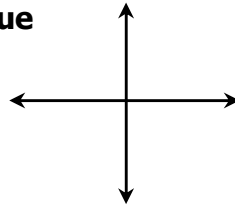
Topic 11: Parent Functions, Transformations, and Graphing

Identify each parent function, then sketch the graph.

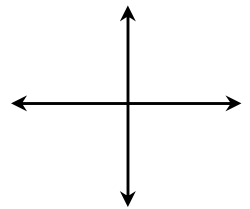
Linear



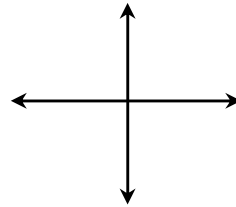
Absolute Value



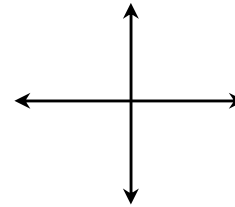
Quadratic



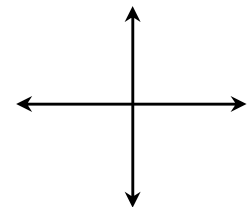
Cubic



Square Root



Cube Root



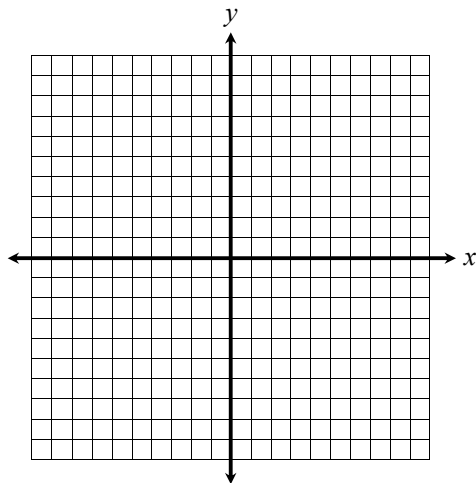
For each equation: (1) identify the function family, and (2) describe the transformations.

70. $f(x) = -2\sqrt{x-9}$

71. $f(x) = \frac{1}{3}(x+7)^2 - 2$

Graph each function and identify its key characteristics.

72. $f(x) = (x-5)^2 - 3$



Domain: _____

Range: _____

Vertex: _____

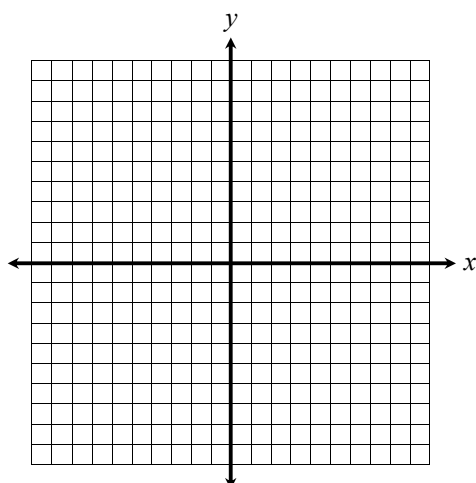
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Inc. Intervals: _____

Dec. Intervals: _____

73. $f(x) = -2|x+3| - 1$



Domain: _____

Range: _____

Vertex: _____

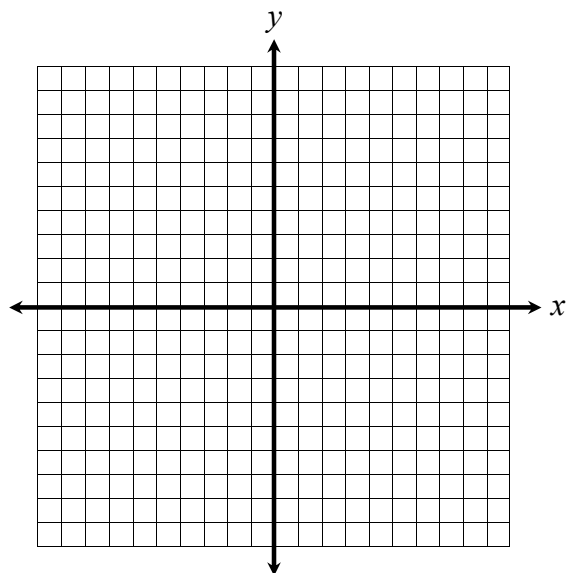
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Inc. Intervals: _____

Dec. Intervals: _____

74. $f(x) = x^3 - 9x^2 + 24x - 16$



Domain: _____

Range: _____

Rel. Maximum(s): _____

Rel. Minimum(s): _____

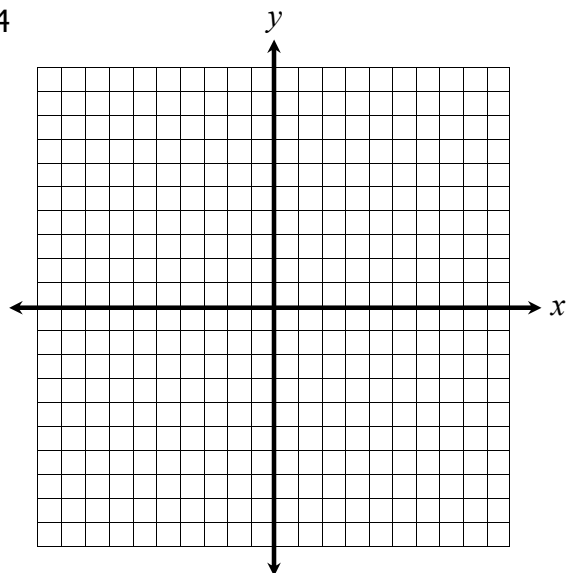
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Inc. Intervals: _____

Dec. Intervals: _____

75. $f(x) = \frac{1}{2}\sqrt{x} + 4$



Domain: _____

Range: _____

Endpoint/Turning Point: _____

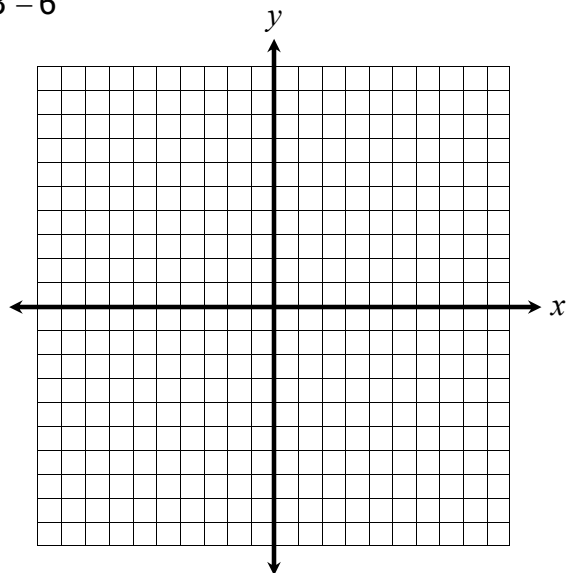
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow 0$, $f(x) \rightarrow$ _____

Inc. Intervals: _____

Dec. Intervals: _____

76. $f(x) = -\sqrt[3]{x+3} - 6$



Domain: _____

Range: _____

Endpoint/Turning Point: _____

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

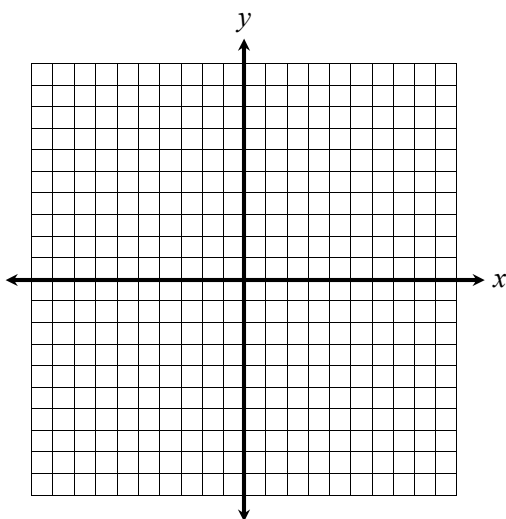
Inc. Intervals: _____

Dec. Intervals: _____

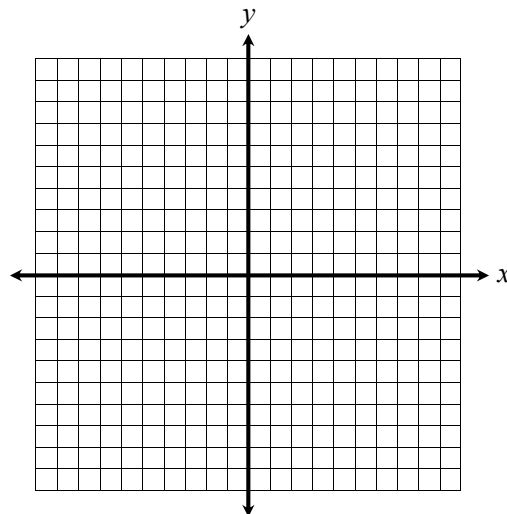
Topic 12: Graphing Absolute Value Inequalities

Graph each inequality.

77. $f(x) < |x - 4| - 1$



78. $f(x) \geq -\frac{1}{2}|x| + 5$



Topic 13: Applications & Regression

79. This past Saturday, Jack worked for 3 hours mowing the lawn and doing chores in the house. Mowing the lawn burns 352 calories per hour and doing chores burns 136 calories per hour. If he burned a total of 570 calories, how long did it take him to mow the lawn?

80. The pilot in a helicopter tossed a life float down to a swimmer in the water. The height, h , of the float after time t seconds is represented by the equation $h(t) = -16t^2 - 3t + 55$. Find the time it takes the float to reach the swimmer.

- 81.** The table below shows the number of teachers, t , and the number of students, s , at 6 high schools in the city. Write a **linear** equation to model the data, then predict the number of teachers for a school with 1,500 students.

t	115	140	76	168	110	92
s	1,122	1,390	788	1,564	1,081	924

- 82.** The table below shows the population, p , of a city during certain years, t . Write a **cubic** equation to model the data, then predict the population of the city in 2020.

t	1993	1996	2000	2005	2009	2014
s	2,850	3,257	4,234	6,475	10,158	27,385

Topic 14: Functions

Use $f(x) = x^2 + 5x - 14$, $g(x) = x + 1$, and $h(x) = 15 - 2x^2$ to answer questions 83 – 88.

83. Find $[f(7)] - 2$

84. If $g(x) = -23$, find x .

85. Find $(f - g)(x)$

86. Find $(f \cdot h)(x)$

87. Find $\left(\frac{g}{f}\right)(x)$

88. Find $(h \circ g)(x)$

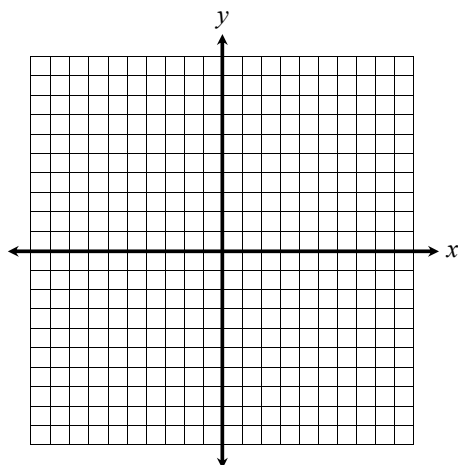
Use $f(x) = x^2 + 5x - 14$, $g(x) = x + 1$, and $h(x) = 15 - 2x^2$ to answer questions 89 – 90.

89. Find $(f + h)(-2)$

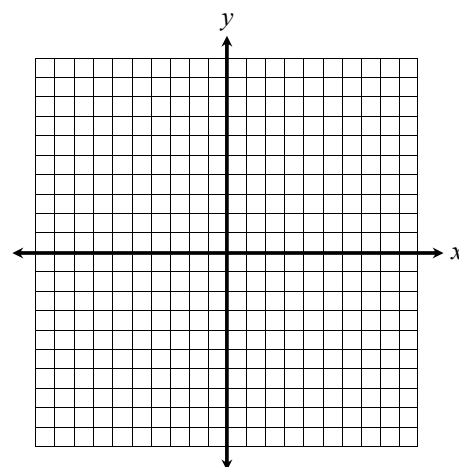
90. Find $g(f(8))$

Find the inverse of each function. Graph to show the relationship.

91. $f(x) = 4 - 2x$



92. $f(x) = \frac{2}{5}x - 1$



Determine whether the pair of functions are inverse functions. Prove algebraically.

93. $f(x) = -6x - 14$ and $g(x) = -\frac{1}{6}x - \frac{7}{3}$

94. $f(x) = \sqrt[3]{x + 8}$ and $g(x) = (x - 8)^3$

Name: _____ Date: _____ Per: _____

Algebra 2: Mid-Year Test

- | | | | | |
|-----------|-----------|-----------|-----------|-----------|
| 1. _____ | 13. _____ | 25. _____ | 37. _____ | 49. _____ |
| 2. _____ | 14. _____ | 26. _____ | 38. _____ | 50. _____ |
| 3. _____ | 15. _____ | 27. _____ | 39. _____ | 51. _____ |
| 4. _____ | 16. _____ | 28. _____ | 40. _____ | 52. _____ |
| 5. _____ | 17. _____ | 29. _____ | 41. _____ | 53. _____ |
| 6. _____ | 18. _____ | 30. _____ | 42. _____ | 54. _____ |
| 7. _____ | 19. _____ | 31. _____ | 43. _____ | 55. _____ |
| 8. _____ | 20. _____ | 32. _____ | 44. _____ | 56. _____ |
| 9. _____ | 21. _____ | 33. _____ | 45. _____ | 57. _____ |
| 10. _____ | 22. _____ | 34. _____ | 46. _____ | 58. _____ |
| 11. _____ | 23. _____ | 35. _____ | 47. _____ | 59. _____ |
| 12. _____ | 24. _____ | 36. _____ | 48. _____ | 60. _____ |

Check All Answers!

Name: _____

Algebra 2

Date: _____ Bell: _____

Mid-Year Test**SHOW ALL WORK NEEDED TO ANSWER EACH QUESTION! Good Luck! ☺**

1. Simplify completely: $\sqrt[3]{320x^{12}y^5}$ A. $8x^9y^2\sqrt[3]{5y^2}$ B. $8x^4y\sqrt[3]{5y^2}$ C. $4x^9y^2\sqrt[3]{5y^2}$ D. $4x^4y\sqrt[3]{5y^2}$	2. Simplify completely: $\sqrt{98m^5} + 4\sqrt{18m^5}$ A. $19m^2\sqrt{2m}$ B. $19m^4\sqrt{2m}$ C. $8m^2\sqrt{29m}$ D. $8m^5\sqrt{29}$
3. Simplify completely: $\sqrt[4]{14k^3} \cdot \sqrt[4]{8k^5}$ A. $2k^2\sqrt[4]{7}$ B. $2k^4\sqrt[4]{7}$ C. $2k^3\sqrt[4]{7k^3}$ D. $4k^6\sqrt[4]{7k^3}$	4. Simplify completely: $(2 + \sqrt{10})(7 + \sqrt{10})$ A. $14 + 2\sqrt{10}$ B. $24 + 9\sqrt{10}$ C. $14 + 9\sqrt{10}$ D. $114 + 9\sqrt{10}$
5. Simplify completely: $\frac{\sqrt[3]{-270p^{14}}}{\sqrt[3]{2p^2}}$ A. $-3p^4\sqrt[3]{10}$ B. $-3p^3\sqrt[3]{10p}$ C. $-3p^4\sqrt[3]{5}$ D. $-3p^3\sqrt[3]{5p}$	6. Simplify completely: $\sqrt{\frac{2x}{27}}$ A. $\frac{\sqrt{2x}}{3}$ B. $\frac{\sqrt{2x}}{9}$ C. $\frac{\sqrt{6x}}{27}$ D. $\frac{\sqrt{6x}}{9}$

7. Simplify completely: $\frac{6}{5 - \sqrt{7}}$ A. $\frac{5 + \sqrt{7}}{3}$ C. $\frac{5 + \sqrt{7}}{18}$ B. $\frac{5 + \sqrt{7}}{6}$ D. $15 + 3\sqrt{7}$	8. Which expression is equivalent to $\sqrt[3]{8x^{12}y^5}$? A. $2x^{\frac{1}{4}}y^{\frac{3}{5}}$ C. $\frac{3}{8}x^{\frac{1}{4}}y^{\frac{3}{5}}$ B. $2x^4y^{\frac{5}{3}}$ D. $\frac{3}{8}x^4y^{\frac{5}{3}}$
9. Which expression is equivalent to $\left(m^{\frac{16}{3}}\right)^{\frac{1}{4}}$? A. $\sqrt[4]{m^3}$ B. $m\sqrt[4]{m^3}$ C. $m\sqrt[3]{m}$ D. $m^2\sqrt[3]{m}$	10. Which expression is equivalent to $\frac{9^{\frac{5}{2}}}{9^4}$? A. 27 C. $\frac{1}{27}$ B. 3 D. $\frac{1}{3}$
11. Which expression is equivalent to $4\sqrt{-36}$? A. $72i$ B. $24i$ C. $12i$ D. $144i$	12. Which expression is equivalent to $-13i^{75}$? A. 13 B. -13 C. $13i$ D. $-13i$
13. Simplify the expression below: $(7 - 2i)^2 - (25 - 12i)$ A. $28 + 21i$ B. $28 - 16i$ C. $20 - 40i$ D. $20 - 16i$	14. Simplify completely: $\frac{5}{2 - i}$ A. $2 + i$ C. $2 - i$ B. $\frac{10 + 5i}{3}$ D. $\frac{10 - 5i}{3}$

15. Which set of numbers does not contain the value below? $(i\sqrt{3})^2$ A. Complex Numbers B. Pure Imaginary Numbers C. Rational Numbers D. Integers	16. Which statement is false? A. All real numbers are complex numbers. B. All irrational numbers are real numbers. C. All complex numbers are pure imaginary numbers. D. All whole numbers are rational numbers.
17. Which statement illustrates the associative property of addition? A. $(1 + \sqrt{3})^2 + \sqrt{7} = 1 + (\sqrt{3} + \sqrt{7})^2$ B. $(-2 + 7i) + 4i = -2 + (7i + 4i)$ C. $\sqrt{18} + 5\sqrt{3} = \sqrt{3}(\sqrt{6} + 5)$ D. $(5i \cdot 2i) + 11i = 11i + (5i \cdot 2i)$	18. Identify the multiplicative inverse of the value below. $\left(-\frac{2i}{9}\right)$ A. $\frac{2i}{9}$ C. $-\frac{9i}{2}$ B. $\frac{9}{2i}$ D. $-\frac{9}{2i}$
19. Which is the complete factorization of the expression $81w^8 - 36w^4$? A. $9w^4(9w^4 - 4)$ B. $9w^4(3w^2 + 2)(3w^2 - 2)$ C. $(9w^4 + 6w^2)(9w^4 - 6w^2)$ D. $9w^4(9w + 1)(w - 4)$	20. Which of the following is the factored form of $x^3 + 343$? A. $(x + 7)^3$ B. $(x + 7)(x^2 + 49)$ C. $(x + 7)(x^2 + 7x + 49)$ D. $(x + 7)(x^2 - 7x + 49)$
21. For which polynomial is the expression $(2y - 3)$ a factor? A. $12y - 8$ B. $45y^2 - 20$ C. $16y^3 - 54$ D. $2y^2 + y - 3$	22. Which of the following expressions is not a factor of $12m^5 - 9m^3 - 3m$? A. $3m$ B. $(m^2 + 1)$ C. $(4m^2 + 1)$ D. $(m - 1)$

23. Find the quotient: $(x^3 - 9x^2 + 5x - 4) \div (x - 2)$

A. $x^2 - 7x - 9 - \frac{22}{x-2}$

B. $x^2 - 11x + 27 - \frac{58}{x-2}$

C. $x^2 - 7x + 19 - \frac{42}{x-2}$

D. $x^2 - 11x - 17 - \frac{38}{x-2}$

24. Solve: $8|3x - 5| = 56$

A. $\{-4\}$

B. $\left\{-4, \frac{2}{3}\right\}$

C. $\left\{-\frac{2}{3}, 4\right\}$

D. $\left\{4, -\frac{14}{3}\right\}$

25. What of the following represents the solution to the inequality below?

$$|x - 7| + 12 \leq 25$$

A. $(-\infty, -20]$ or $[6, \infty)$

B. $(-\infty, -6]$ or $[20, \infty)$

C. $[-20, 6]$

D. $[-6, 20]$

26. Which of the following describes the roots of the function $f(x) = -x^2 - 3x + 1$?

A. exactly 1 rational root

B. 2 distinct rational roots

C. 2 distinct irrational roots

D. 2 distinct imaginary roots

27. Solve: $-2x^2 + 12 = 68$

A. $\{\pm 2i\sqrt{7}\}$

B. $\{\pm 2i\sqrt{10}\}$

C. $\{\pm 2\sqrt{7}\}$

D. $\{\pm 2\sqrt{10}\}$

28. Solve: $x^2 - 12x - 14 = 0$

- A. $\{-6 \pm i\sqrt{22}\}$
- B. $\{6 \pm i\sqrt{22}\}$
- C. $\{-6 \pm 5\sqrt{2}\}$
- D. $\{6 \pm 5\sqrt{2}\}$

29. Which equation has a solution set of

$$\left\{-\frac{3}{2}, 4\right\}?$$

- A. $2x^2 = 12 - 5x$
- B. $3x^2 - 12 = x^2 + 5x$
- C. $3x^2 - 9x = x + 8$
- D. $2x^2 + 10x + 3 = 11 - x^2$

30. Which is a solution to the equation $x^3 - 64 = 0$?

- A. $x = -4$
- B. $x = -8$
- C. $x = -2 + 2i\sqrt{3}$
- D. $x = -2 + 2\sqrt{3}$

31. Which is not a solution to the equation below?

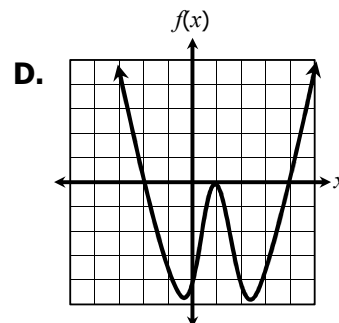
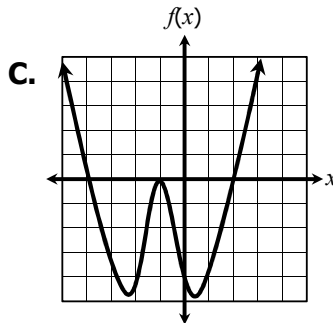
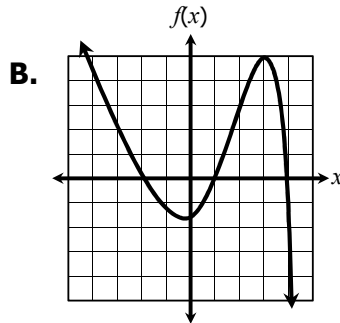
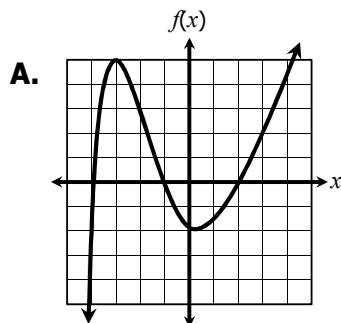
$$x^3 + 7x^2 - 9x - 63 = 0$$

- A. 9
- B. -7
- C. 3
- D. -3

32. A polynomial function has a zero at $x = 4i$. Which of the following expressions must be a factor of the polynomial?

- A. $(x^2 + 8)$
- B. $(x^2 - 8)$
- C. $(x^2 + 16)$
- D. $(x^2 - 16)$

33. A polynomial function has a zeros at -4 (multiplicity 3), -1 (multiplicity 8) and 2 (multiplicity 1). Which equation could represent this function?



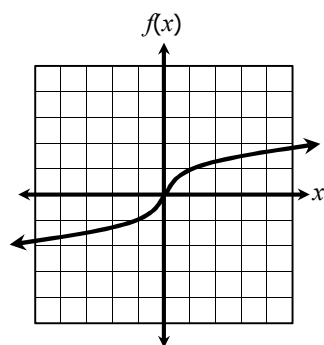
34. Solve: $\sqrt{12 - 2x} + 3 = 7$

- A. $x = -17$
- B. $x = -2$
- C. $x = -14$
- D. $x = 16$

35. Solve: $\sqrt[4]{\frac{1}{3}x - 8} = 2$

- A. $x = 72$
- B. $x = 8$
- C. $x = \frac{10}{3}$
- D. $x = -2$

36. The graph of a parent function is shown below. Which function belongs to the same family?

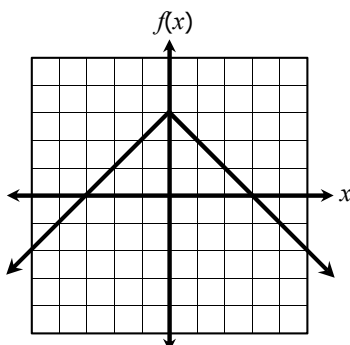


- A. $f(x) = \sqrt{x} + 6$
- B. $f(x) = \sqrt[3]{x - 4}$
- C. $f(x) = 2x^3 + 5$
- D. $f(x) = (x - 1)^2$

37. The graph of the absolute value parent function is vertically stretched, then shifted 3 units left, 8 units down. Which equation could represent this function?

- A. $f(x) = 2|x + 3| - 8$
- B. $f(x) = \frac{1}{2}|x + 3| - 8$
- C. $f(x) = 2|x - 3| - 8$
- D. $f(x) = \frac{1}{2}|x - 3| - 8$

38. Which equation defines the function graphed below?



- A. $f(x) = -|x + 3|$
- B. $f(x) = -|x - 3|$
- C. $f(x) = -|x| + 3$
- D. $f(x) = -|x| - 3$

39. Which function has two distinct real zeros?

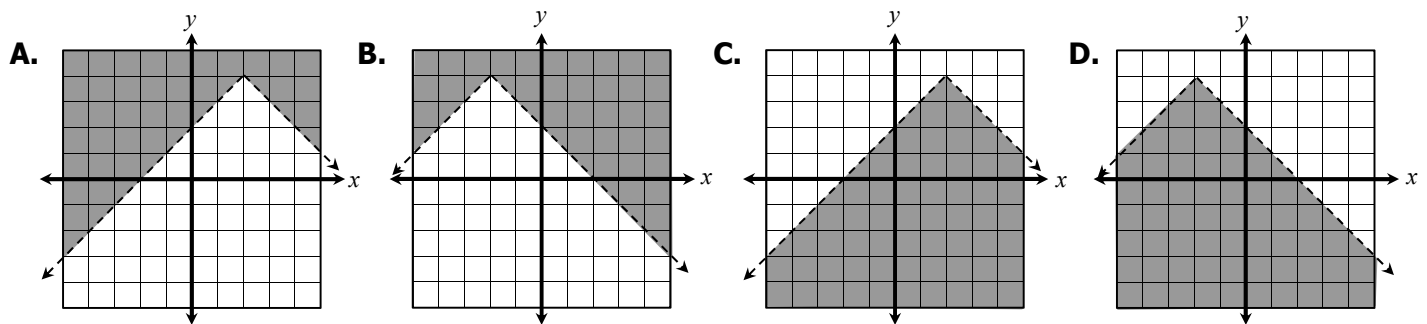
- A. $f(x) = 2x + 2$
- B. $f(x) = (x - 2)^2$
- C. $f(x) = x^3 + 2$
- D. $f(x) = |x| - 2$

40. Which of the following describes the range of the function below? $f(x) = -(x + 3)^2 - 5$ A. $\{y \mid y \leq 3\}$ B. $\{y \mid y \geq 3\}$ C. $\{y \mid y \leq -5\}$ D. $\{y \mid y \geq -5\}$	41. Which function does not have a domain of all real numbers? A. $f(x) = \sqrt{x} - 1$ B. $f(x) = -x^3 + 4x$ C. $f(x) = - x - 5 + 1$ D. $f(x) = \frac{1}{2}\sqrt[3]{x} + 6$
42. Identify the vertex of the function below. $f(x) - 4 = (x + 1)^2$ A. $(-4, 1)$ B. $(1, -4)$ C. $(-1, -4)$ D. $(-1, 4)$	43. Which of the following statements is true regarding the function below? $f(x) = (x - 3)(x - 7)^2$ A. $(3, 0)$ is a zero and a relative minimum B. $(3, 0)$ is a zero and a relative maximum C. $(7, 0)$ is a zero and a relative minimum D. $(7, 0)$ is a zero and a relative maximum
44. The graph of the cubic parent function is reflected across the x-axis. Which of the following describes the end behavior of the new function as x approaches negative infinity? A. $f(x)$ approaches negative infinity B. $f(x)$ approaches positive infinity C. $f(x)$ approaches 0 D. $f(x)$ approaches 3	
45. Which function family can have more than one increasing interval? A. Square Root B. Cube Root C. Quadratic D. Cubic	46. During which interval is the function below only decreasing? $f(x) = (x + 2)(x - 1)^2$ A. $-1 < x < 1$ B. $-2 < x < 1$ C. $-1 < x < 2$ D. $-\infty < x < 1$

47. The graph of a quadratic function increases through the interval $-\infty < x < 5$, then decreases through the interval $5 < x < \infty$. If the vertex of the graph is located at $(5, 9)$, which equation could represent this function?

- A. $f(x) = (x + 5)^2 + 9$
- B. $f(x) = (x - 5)^2 + 9$
- C. $f(x) = -(x + 5)^2 + 9$
- D. $f(x) = -(x - 5)^2 + 9$

48. Which graph represents the inequality $f(x) > -|x - 2| + 4$?



49. Kate's soccer team is selling coupon books and recipe books to raise money for transportation to their games. Coupon books cost \$12 each and recipe books cost \$7.50 each. If Kate sold a total of 42 books and raised \$441, how many coupon books did she sell?

- A. 18 coupon books
- B. 24 coupon books
- C. 28 coupon books
- D. 14 coupon books

50. A baseball is thrown from a height of 5 feet. The height, h , of the ball at time t seconds is modeled by the equation $h(t) = -16t^2 + 100t + 5$. How long will it take the ball to reach the ground?

- A. 5.8 seconds
- B. 6.3 seconds
- C. 7.2 seconds
- D. 7.6 seconds

51. The table below shows the yearly average selling price of a home in the Orchard Woods subdivision during certain years. Using a quartic model, approximate the selling price of a home in Orchard Woods in 2020.

Year	Average Selling Price
1998	\$144,000
2003	\$174,600
2010	\$175,120
2012	\$180,500
2015	\$203,000

- A.** \$300,303
- B.** \$306,128
- C.** \$310,502
- D.** \$315,457

Use the functions $f(x) = 4x + 1$, $g(x) = 2x^2 - 6x$, and $h(x) = -x^2 + 3x - 19$ for questions 52-57.

52. Find $-2 \cdot [g(-2)]$

- A.** 8
- B.** -40
- C.** 20
- D.** -16

53. If $f(x) = -13$, find x .

- A.** -3
- B.** $-\frac{2}{7}$
- C.** -51
- D.** $-\frac{7}{2}$

54. Find $(h - g)(x)$

- A.** $x^2 - 9x - 19$
- B.** $3x^2 - 9x + 19$
- C.** $-3x^2 - 3x - 19$
- D.** $-3x^2 + 9x - 19$

55. Find $(f \cdot h)(x)$

- A.** $-x^2 + 12x + 3x - 18$
- B.** $-4x^3 + 6x^2 - 11x - 18$
- C.** $-4x^3 + 11x^2 - 73x - 19$
- D.** $-4x^3 + 12x^2 - 76x - 19$

56. Which describes the domain of the function $\left(\frac{g}{f}\right)(x)$?

- A.** All Real Numbers
- B.** $\left\{x \mid x \neq \frac{1}{4}\right\}$
- C.** $\{x \mid x \neq -1\}$
- D.** $\left\{x \mid x \neq -\frac{1}{4}\right\}$

57. Find $(g \circ f)(5)$

- A.** 81
- B.** 420
- C.** 756
- D.** 915

58. Find the inverse of the function below.

$$f(x) = 8x - 4$$

A. $f^{-1}(x) = 2x + 1$

C. $f^{-1}(x) = 2x - 1$

B. $f^{-1}(x) = \frac{1}{8}x + \frac{1}{2}$

D. $f^{-1}(x) = \frac{1}{8}x - \frac{1}{2}$

59. The point $(-7, 1)$ lies on the graph of $g(x)$. Which point must lie on the graph of $g^{-1}(x)$?

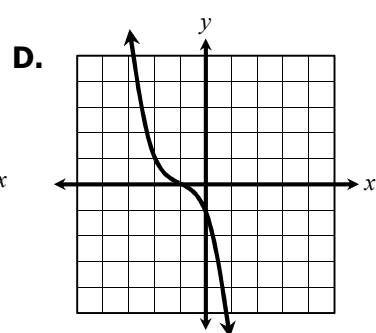
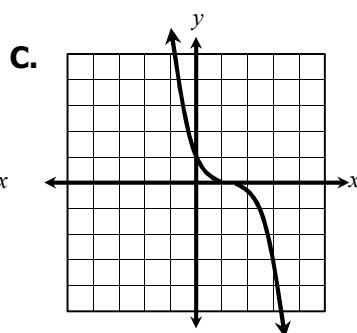
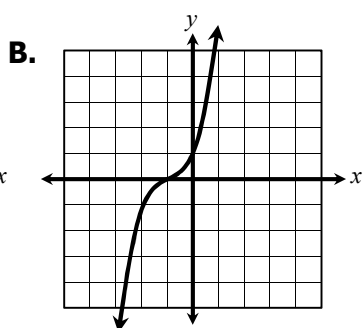
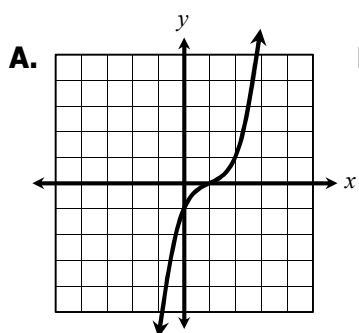
A. $(7, -1)$

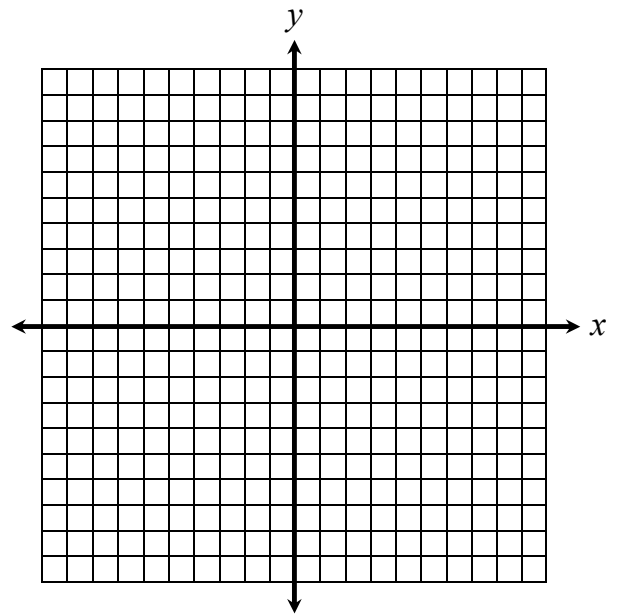
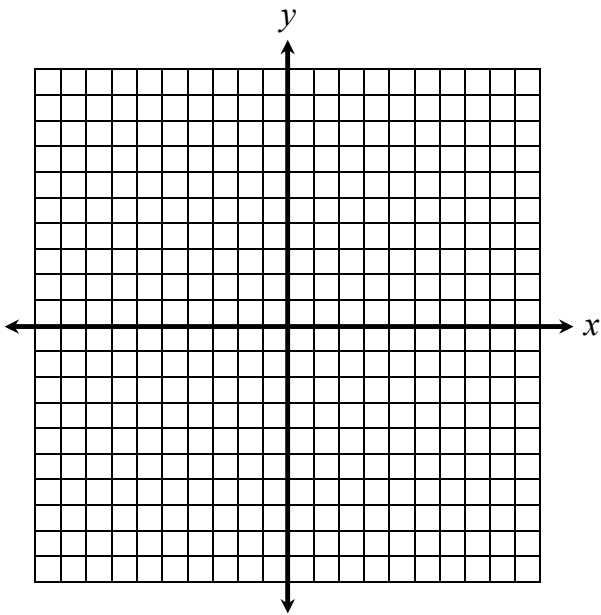
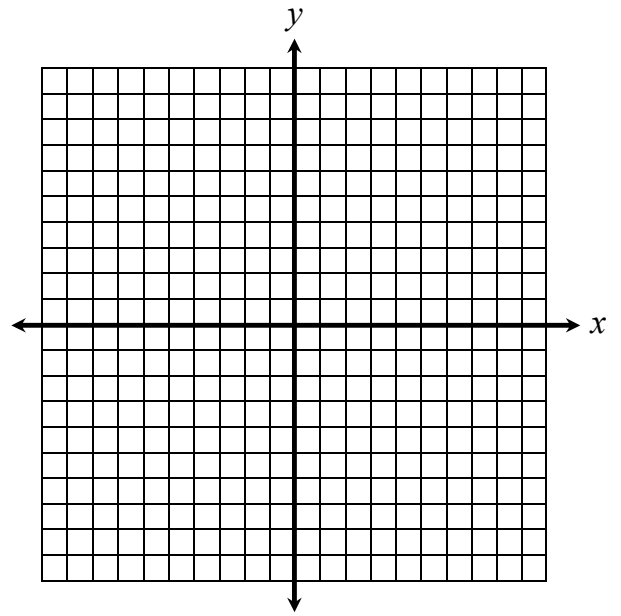
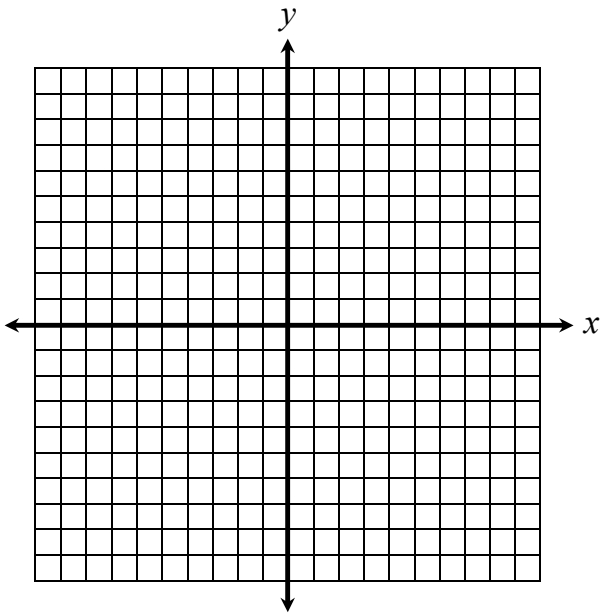
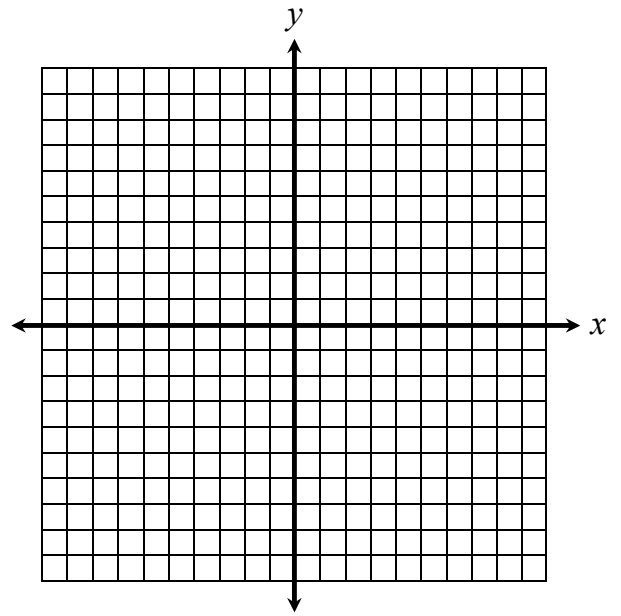
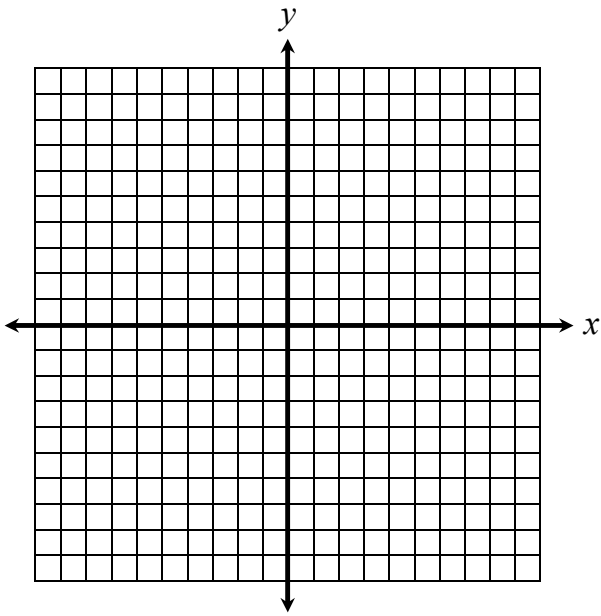
B. $(1, -7)$

C. $(-1, 7)$

D. $(-7, -1)$

60. Given $f(x) = \sqrt[3]{x} + 1$, which graph represents $f^{-1}(x)$?





Algebra 2 Mid-Year Test Study Guide

Name: Key!

Date: _____ Block: _____

Topic 1: Radicals

Simplify each expression.

$$\begin{aligned} 1. \sqrt{243a^{25}} \\ = \sqrt{81a^{24}} \cdot \sqrt{3a} \\ = \boxed{9a^{12}\sqrt{3a}} \end{aligned}$$

$$\begin{aligned} 2. \sqrt[3]{-18p^9} \cdot \sqrt[3]{4p^5} \\ = \sqrt[3]{-72p^{14}} \\ = \sqrt[3]{-8p^{12}} \cdot \sqrt[3]{9p^2} \\ = \boxed{-2p^4\sqrt[3]{9p^2}} \end{aligned}$$

$$\begin{aligned} 3. \frac{\sqrt[4]{32a^{13}b^5}}{\sqrt[4]{2a^2b}} &= \sqrt[4]{16a^{11}b^4} \\ &= \sqrt[4]{16a^8b^4} \cdot \sqrt[4]{a^3} \\ &= \boxed{2a^2b\sqrt[4]{a^3}} \end{aligned}$$

$$\begin{aligned} 4. 3\sqrt{72} - 2\sqrt{8} + 2\sqrt{20} \\ = 3\sqrt{36} \sqrt{2} - 2\sqrt{4} \sqrt{2} + 2\sqrt{4} \sqrt{5} \\ = 3 \cdot 6 \sqrt{2} - 2 \cdot 2 \sqrt{2} + 2 \cdot 2 \sqrt{5} \\ = 18\sqrt{2} - 4\sqrt{2} + 4\sqrt{5} \\ = \boxed{14\sqrt{2} + 4\sqrt{5}} \end{aligned}$$

$$\begin{aligned} 5. \sqrt{2}(\sqrt{2} - 6\sqrt{10}) \\ = \sqrt{4} - 6\sqrt{20} \\ = 2 - 6\sqrt{4} \sqrt{5} \\ = 2 - 6 \cdot 2\sqrt{5} = \boxed{2 - 12\sqrt{5}} \end{aligned}$$

$$\begin{aligned} 6. \frac{6}{\sqrt{12}} \frac{\sqrt{3}}{\sqrt{3}} &= \frac{6\sqrt{3}}{\sqrt{36}} \\ &= \frac{6\sqrt{3}}{6} \\ &= \boxed{\sqrt{3}} \end{aligned}$$

$$\begin{aligned} 7. \frac{(\sqrt{2} - 2\sqrt{3})\sqrt{8}}{\sqrt{8} \cdot \sqrt{8}} \\ \frac{\sqrt{16} - 2\sqrt{24}}{\sqrt{64}} = \frac{4 - 2\sqrt{4}\sqrt{6}}{8} = \frac{4 - 4\sqrt{6}}{8} \\ = \boxed{\frac{1 - \sqrt{6}}{2}} \end{aligned}$$

$$\begin{aligned} 8. \frac{(2 + \sqrt{3})(5 + \sqrt{3})}{(5 - \sqrt{3})(5 + \sqrt{3})} \\ = \frac{10 + 2\sqrt{3} + 5\sqrt{3} + \sqrt{9}}{25 + 5\sqrt{3} - 5\sqrt{3} - \sqrt{9}} \\ = \frac{10 + 7\sqrt{3} + 3}{25 - 3} = \boxed{\frac{13 + 7\sqrt{3}}{22}} \end{aligned}$$

Topic 2: Rational Exponents

Write in radical form. Simplify if needed.

$$\begin{aligned} 9. 150^{\frac{1}{2}} &= \sqrt{150} \\ &= \sqrt{25} \sqrt{6} \\ &= \boxed{5\sqrt{6}} \end{aligned}$$

$$\begin{aligned} 10. (-32)^{\frac{1}{3}} &= \sqrt[3]{-32} \\ &= \sqrt[3]{-8} \sqrt[3]{4} \\ &= \boxed{-2\sqrt[3]{4}} \end{aligned}$$

$$\begin{aligned} 11. x^{\frac{11}{4}} y^{\frac{5}{4}} &= \sqrt[4]{x^{11}y^5} \\ &= \sqrt[4]{x^8y^4} \cdot \sqrt[4]{x^3y} \\ &= \boxed{x^2y\sqrt[4]{x^3y}} \end{aligned}$$

Write in exponential form.

$$12. \sqrt[3]{64} = \boxed{(64)^{\frac{1}{3}}}$$

$$13. \sqrt{(xy)^5} = \boxed{(xy)^{\frac{5}{2}}}$$

$$\begin{aligned} 14. \sqrt[4]{10r^7s^{12}} &= 10^{\frac{1}{4}} r^{\frac{7}{4}} s^{\frac{12}{4}} \\ &= \boxed{10^{\frac{1}{4}} r^{\frac{7}{4}} s^3} \end{aligned}$$

Simplify. Write all answers in simplest radical form.

$$15. n^{\frac{7}{2}} \cdot n^{\frac{2}{2}} = n^{\frac{9}{2}}$$

$$= \sqrt{n^9}$$

$$= \sqrt{n^8 \cdot n}$$

$$= [n^4 \sqrt{n}]$$

$$16. \frac{k^{\frac{15}{4}}}{k^{\frac{1}{2}} = \frac{2}{4}} = K^{\frac{13}{4}}$$

$$= \sqrt[4]{K^{13}}$$

$$= \sqrt[4]{K^{12} \cdot K}$$

$$= [K^3 \sqrt[4]{K}]$$

$$17. (8^{\frac{4}{3}})^{\frac{1}{2}} = 8^{-\frac{2}{3}}$$

$$= \frac{1}{\sqrt[3]{64}}$$

$$= [\frac{1}{4}]$$

Topic 3: Complex Numbers

***Recall:** $i = \sqrt{-1}$ $i^2 = -1$ $i^3 = -i$ $i^4 = 1$

Simplify each expression.

$$18. \sqrt{-81}$$

$$\sqrt{-1 \cdot 81}$$

$$[9i]$$

$$19. 2\sqrt{-14} \cdot \sqrt{8}$$

$$2\sqrt{-112}$$

$$2\sqrt{16 \cdot 7}$$

$$[8i\sqrt{7}]$$

$$20. 18i^{19}$$

$$18(i^2)^{24} \cdot i$$

$$18(-1)^{24} \cdot i$$

$$18(1) \cdot i = [18i]$$

$$21. (i^3 \sqrt{5})^2 \cdot 2i$$

$$i^6 \cdot 5 \cdot 2i$$

$$10i^7$$

$$10(i^2)^3 \cdot i$$

$$10(-1)^3 \cdot i = [-10i]$$

$$22. (4-7i)(5+2i)$$

$$[-1-9i]$$

$$23. (9+2i)(4-i)$$

$$36-9i+8i-2i^2$$

$$36-i-2(-1)$$

$$[38-i]$$

$$24. (6-2i)^2$$

$$(6-2i)(6-2i)$$

$$36-12i-12i+4i^2$$

$$36-24i+4(-1)$$

$$[32-24i]$$

$$25. \frac{15 \cdot i}{6i \cdot i} = \frac{15i}{6i^2}$$

$$= \frac{15i}{-6}$$

$$= [-\frac{5i}{2}]$$

$$26. \frac{1-5i \cdot i}{2i \cdot i} = \frac{i-5i^2}{2i^2}$$

$$= \frac{5+i}{-2}$$

$$= [-\frac{5+i}{2}]$$

$$27. \frac{3-3i}{7-i} \cdot \frac{(7+i)}{(7+i)}$$

$$= \frac{21+3i-21i-3i^2}{49+7i-7i-i^2}$$

$$= \frac{24-18i}{50} = [\frac{12-9i}{25}]$$

Topic 4: Classifying Numbers & Identifying Properties

Simplify (if possible), then name all sets to which each value belongs.

$$28. \frac{\sqrt{64}}{5} = \frac{8}{5}$$

C, R, Q

$$29. \sqrt{-36} = 6i$$

C, pure imag.

$$30. \sqrt{3} - \sqrt{3} = 0$$

C, R, Q, Z, W

$$31. |1-4^2| = |-15|$$

$$= 15$$

C, R, Q, Z, W, N

$$32. 2+5i$$

C

$$33. \sqrt{15}$$

C, R, I

$$34. -\sqrt{\frac{45}{5}} = -3$$

C, R, Q, Z

$$35. -8i \cdot 5i = -40i^2$$

$$= -40(-1)$$

$$= 40$$

C, R, Q, Z, W, N

Name the property that justifies each statement.	
36. $\sqrt{5}(8 - \sqrt{3}) = 8\sqrt{5} - \sqrt{3} \cdot \sqrt{5}$ Distributive Property	37. $2m^3 + (-2m^3) = 0$ Inverse (Add.)
38. $-9 \cdot (2 + 3i) = (2 + 3i) \cdot -9$ Commutative Property (Mult.)	39. $\left(\frac{x}{5}\right) \cdot \left(\frac{5}{x}\right) = 1$ Inverse (Mult.)

Topic 5: Factoring		
Identify the special factoring patterns.		
DIFFERENCE OF SQUARES $a^2 - b^2 =$ $(a-b)(a+b)$	SUM OF CUBES $a^3 + b^3 =$ $(a+b)(a^2 - ab + b^2)$	DIFFERENCE OF CUBES $a^3 - b^3 =$ $(a-b)(a^2 + ab + b^2)$
Factor the expressions below completely.		
40. $36x^2 - 64y^2$ $= 4(9x^2 - 16y^2)$ $= 4(3x - 4y)(3x + 4y)$	41. $m^3 + 64$ $= (m+4)(m^2 - 4m + 16)$	
42. $81k^4 - 3k$ $= 3k(27k^3 - 1)$ $= 3k(3k-1)(9k^2 + 3k + 1)$	43. $2xy^5 + 250xy^2$ $= 2xy^2(y^3 + 125)$ $= 2xy^2(y+5)(y^2 - 5y + 25)$	
44. $p^3 - 10p^2 + 25p$ $= p(p^2 - 10p + 25)$ $= p(p-5)(p-5)$ $= p(p-5)^2$	45. $x^4 - 11x^2 + 28$ $= (x^2 - 4)(x^2 - 7)$ $= (x+2)(x-2)(x^2 - 7)$	
46. $4x^3 + 18x^2 - 10x$ $= 2x(2x^2 + 9x - 5)$ $= 2x(2x-1)(x+5)$	47. $9x^3 - 63x^2 - x + 7$ $= 9x^2(x-7) - 1(x-7)$ $= (9x^2 - 1)(x-7)$ $= (3x-1)(3x+1)(x-7)$	

Topic 6: Dividing Polynomials

Find the quotient by long or synthetic division.

48. $(a^3 + 6a^2 + 2a - 20) \div (a + 2)$

$$\begin{array}{r}
 a^2 + 4a - 6 \quad -8 \\
 a+2 \overline{) a^3 + 6a^2 + 2a - 20} \quad a+2 \\
 \underline{-(a^3 + 2a^2)} \downarrow \\
 4a^2 + 2a \\
 \underline{-(4a^2 + 8a)} \downarrow \\
 -6a - 20 \\
 \underline{-(-6a - 12)} \\
 -8
 \end{array}$$

$$\begin{array}{r}
 -2 \overline{) 1 \ 6 \ 2 \ -20} \\
 \underline{ \downarrow -2 \ -8 \ 12} \\
 1 \ 4 \ -6 \ \textcircled{-8}
 \end{array}$$

$$a^2 + 4a - 6 - \frac{8}{a+2}$$

49. $(x^4 - 64x^2 - 7x + 61) \div (x - 8) + 0x - 7 + \frac{5}{x-8}$

$$\begin{array}{r}
 x^3 + 8x^2 + 0x - 7 \quad + \frac{5}{x-8} \\
 x-8 \overline{) x^4 + 0x^3 - 64x^2 - 7x + 61} \\
 \underline{-(x^4 - 8x^3)} \\
 8x^3 - 64x^2 \\
 \underline{-(8x^3 - 64x^2)} \\
 0x^2 - 7x \\
 \underline{-(0x^2 + 0x)} \\
 -7x + 61 \\
 \underline{-(-7x + 56)} \\
 5
 \end{array}$$

$$\begin{array}{r}
 8 \overline{) 1 \ 0 \ -64 \ -7 \ 61} \\
 \underline{ \downarrow 8 \ 64 \ 0 \ -56} \\
 1 \ 8 \ 0 \ -7 \ \textcircled{5}
 \end{array}$$

$$x^3 + 8x^2 - 7 + \frac{5}{x-8}$$

Topic 7: Absolute Value Equations & Inequalities

Solve each equation. Check all solutions.

50. $|4m + 5| = 9$

$$4m + 5 = 9$$

$$4m = 4$$

$$m = 1 \quad \checkmark$$

$$4m + 5 = -9$$

$$4m = -14$$

$$m = -\frac{7}{2} \quad \checkmark$$

$$m = \left\{ -\frac{7}{2}, 1 \right\}$$

$$|4(1) + 5| = 9$$

$$|9| = 9$$

$$|4(-\frac{7}{2}) + 5| = 9$$

$$|-14 + 5| = 9$$

$$|-9| = 9$$

51. $-5|7 - x| + 6 = -14$

$$-5|7 - x| = -20$$

$$|7 - x| = 4$$

$$7 - x = 4$$

$$-x = -3$$

$$x = 3 \quad \checkmark$$

$$7 - x = -4$$

$$-x = -11$$

$$x = 11 \quad \checkmark$$

$$m = \{3, 11\}$$

$$-5|7 - 3| + 6 = -14$$

$$-5(4) + 6 = -14$$

$$-14 = -14$$

$$-5|7 - 11| + 6 = -14$$

$$-5(4) + 6 = -14$$

$$-14 = -14$$

Solve each inequality and graph your solution. Write the solution in interval notation.

52. $|x - 2| \leq 3$

$$x - 2 \leq 3$$

$$x - 2 \geq -3$$

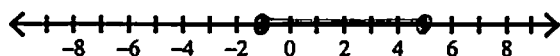
$$x \leq 5$$

and

$$x \geq -1$$

$$-1 \leq x \leq 5$$

$$[-1, 5]$$



Interval Notation:

53. $-2|2x - 1| < -14$

$$|2x - 1| > 7$$

$$2x - 1 > 7$$

$$2x - 1 < -7$$

$$2x > 8$$

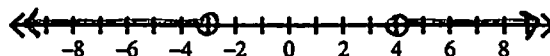
$$2x < -6$$

$$x > 4$$

or

$$x < -3$$

$$(-\infty, -3) \cup (4, \infty)$$



Interval Notation:

Topic 8: Quadratic Equations

Discriminant Formula:

- If $d < 0$, there are 2 imaginary roots.
- If $d = 0$, there is 1 rational root
- If $d > 0$ and a perfect square, there are 2 rational roots
- If $d > 0$ and not a perfect square, there are 2 irrational roots.

Find the discriminant of each equation. Then, determine the number and type of roots.

54. $x^2 - 7x + 15 = 0$

$$= (-7)^2 - 4(1)(15)$$

$$= 49 - 60$$

$$= -11 ; 2 \text{ imaginary roots}$$

55. $2x^2 - 72 = 0$

$$= (0)^2 - 4(2)(-72)$$

$$= 0 + 576$$

$$= 576 ; 2 \text{ rational roots}$$

56. $-x^2 + 9x + 23 = 0$

$$= (9)^2 - 4(-1)(23)$$

$$= 81 + 92$$

$$= 173 ; 2 \text{ irrational roots}$$

Solve using the most appropriate method. Simplify all irrational and complex solutions.

57. $2x^2 - 18 = 78$

$$2x^2 = 96$$

$$x^2 = 48$$

$$x = \pm \sqrt{48}$$

$$x = \pm \sqrt{16} \sqrt{3}$$

$$x = \{\pm 4\sqrt{3}\}$$

* No b-term,
use sq. roots *

58. $x^2 - 16x + 73 = 0$

$$x^2 - 16x = -73$$

$$x^2 - 16x + 64 = -73 + 64$$

$$(x - 8)^2 = -9$$

$$x - 8 = \pm \sqrt{-9}$$

$$x - 8 = \pm 3i$$

$$x = \{8 \pm 3i\}$$

59. $9x^2 - 18x = 11$

$$9x^2 - 18x - 11 = 0$$

$$x = \frac{18 \pm \sqrt{(-18)^2 - 4(9)(-11)}}{2(9)}$$

$$x = \frac{18 \pm \sqrt{324 + 396}}{18}$$

$$x = \frac{18 \pm \sqrt{720}}{18}$$

$$x = \frac{18 \pm 12\sqrt{5}}{18}$$

$$x = \left\{ \frac{3 \pm 2\sqrt{5}}{3} \right\}$$

60. $2x^2 - 10x - 26 = x - 5$

$$2x^2 - 11x - 21 = 0$$

$$(2x + 3)(x - 7) = 0$$

$$2x + 3 = 0$$

$$x - 7 = 0$$

$$2x = -3$$

$$x = 7$$

$$x = -\frac{3}{2}$$

$$x = \left\{ -\frac{3}{2}, 7 \right\}$$

61. The roots of a quadratic equation are -7 and 1. Write an equation that could represent this function. Give your answer in standard form and factored form.

$$(x + 7)(x - 1) = 0$$

$$x^2 + 6x - 7 = 0$$

Topic 9: Polynomial Equations

Solve each equation. Simplify all irrational and complex solutions.

62. $x^3 - 27 = 0$

$$(x - 3)(x^2 + 3x + 9) = 0$$

$$x - 3 = 0$$

$$x = 3$$

$$x = \frac{-3 \pm \sqrt{(3)^2 - 4(1)(9)}}{2(1)}$$

$$x = \frac{-3 \pm \sqrt{-27}}{2}$$

$$x = \frac{-3 \pm 3i\sqrt{3}}{2}$$

$$x = \left\{ 3, \frac{-3 \pm 3i\sqrt{3}}{2} \right\}$$

63. $2x^3 - 5x^2 + 18x - 45 = 0$

$$x^2(2x - 5) + 9(2x - 5) = 0$$

$$(x^2 + 9)(2x - 5) = 0$$

$$x^2 + 9 = 0$$

$$2x - 5 = 0$$

$$x^2 = -9$$

$$2x = 5$$

$$x = \pm 3i$$

$$x = \frac{5}{2}$$

64. A polynomial function is defined by the equation below. Identify the zeros, their multiplicity, and the effect on the graph

$$f(x) = -2x(3x+2)^7(x-1)^2$$

Zero	Multiplicity	Effect
0	1	intersect
$-2/3$	7	intersect
1	2	bounce

65. The zeros of a polynomial equation are -4 (multiplicity 2) and 3 (multiplicity 1). Write an equation in standard form that could represent the function.

$$f(x) = (x+4)^2(x-3)$$

$$f(x) = (x^2+8x+16)(x-3)$$

$$f(x) = x^3 - 3x^2 + 8x^2 - 24x + 16x - 48$$

$$f(x) = x^3 + 5x^2 - 8x - 48$$

Topic 10: Radical Equations

Solve each equation. Check all solutions.

66. $\sqrt{8x-7}+10=15$

$$\sqrt{8x-7}=5$$

$$8x-7=25$$

$$8x=32$$

$$\boxed{x=4} \checkmark$$

ck
 $\sqrt{8(4)-7}+10=15$

$$\sqrt{25}+10=15$$

$$5+10=15$$

67. $\sqrt[3]{2x+12}=4$

$$2x+12=64$$

$$2x=52$$

$$\boxed{x=26} \checkmark$$

ck
 $\sqrt[3]{2(26)+12}=4$

$$\sqrt[3]{52+12}=4$$

$$\sqrt[3]{64}=4$$

68. $\sqrt{-7-4m}=\sqrt{2m+11}$

ck
 $-7-4m=2m+11$ $\sqrt{-7+12}=\sqrt{6+11}$

$$-18=6m$$

$$\sqrt{5}=\sqrt{5}$$

$$\boxed{-3=m} \checkmark$$

69. $\sqrt{56-a}=a$

$$56-a=a^2$$

$$0=a^2+a-56$$

$$0=(a+8)(a-7)$$

$$a+8=0$$

$$\boxed{a=-8}$$

$$a-7=0$$

$$\boxed{a=7} \checkmark$$

ck
 $\sqrt{56+8} \neq 8$
 $\sqrt{64} \neq 8$

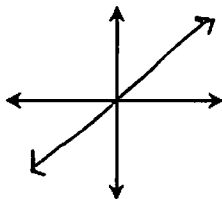
ck
 $\sqrt{56-7}=7$
 $\sqrt{49}=7$

Topic 11: Parent Functions, Transformations, and Graphing

Identify each parent function, then sketch the graph.

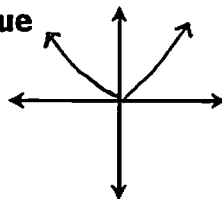
Linear

$$f(x) = x$$



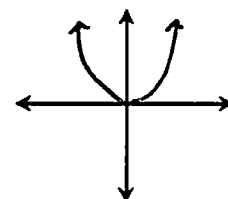
Absolute Value

$$f(x) = |x|$$



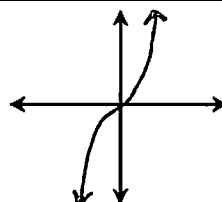
Quadratic

$$f(x) = x^2$$



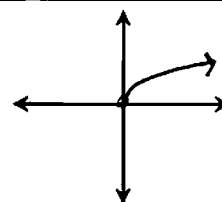
Cubic

$$f(x) = x^3$$



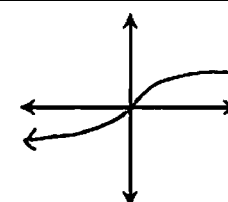
Square Root

$$f(x) = \sqrt{x}$$



Cube Root

$$f(x) = \sqrt[3]{x}$$



For each equation: (1) identify the function family, and (2) describe the transformations.

70. $f(x) = -2\sqrt{x-9}$ Sq. root family

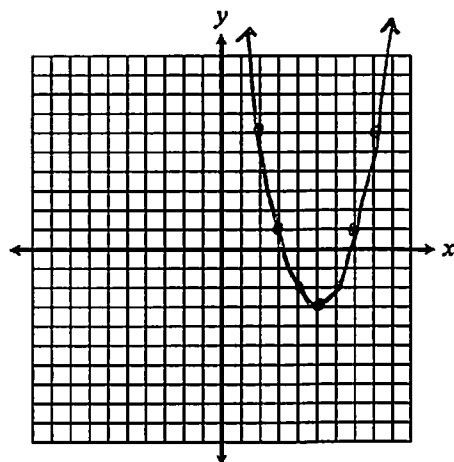
reflect across x-axis
vert. stretch by 2
right 9

71. $f(x) = \frac{1}{3}(x+7)^2 - 2$ quadratic family

vert. compression by $\frac{1}{3}$
left 7
down 2

Graph each function and identify its key characteristics.

72. $f(x) = (x-5)^2 - 3$



Domain: $\mathbb{R}$

Range: $\{y \mid y \geq -3\}$

Vertex: $(5, -3)$

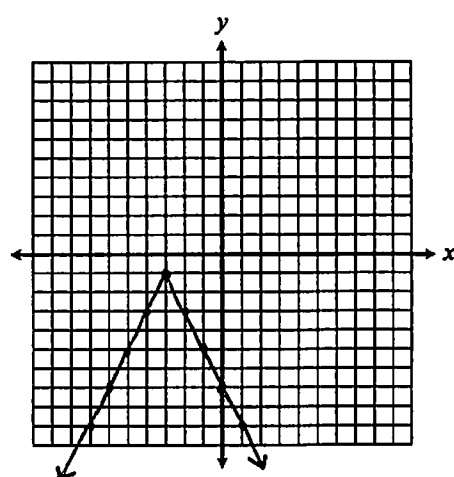
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

Inc. Intervals: $(5, \infty)$

Dec. Intervals: $(-\infty, 5)$

73. $f(x) = -2|x+3| - 1$



Domain: $\mathbb{R}$

Range: $\{y \mid y \leq -1\}$

Vertex: $(-3, -1)$

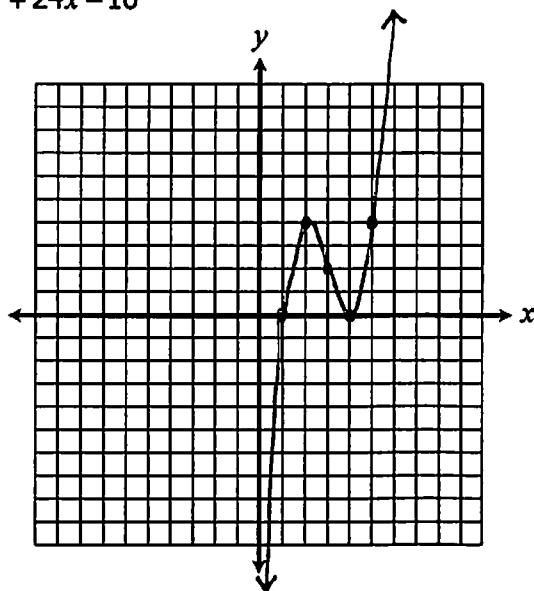
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Inc. Intervals: $(-\infty, -3)$

Dec. Intervals: $(-3, \infty)$

74. $f(x) = x^3 - 9x^2 + 24x - 16$



Domain: $\mathbb{R}$

Range: $\mathbb{R}$

Rel. Maximum(s): $(2, 4)$

Rel. Minimum(s): $(4, 0)$

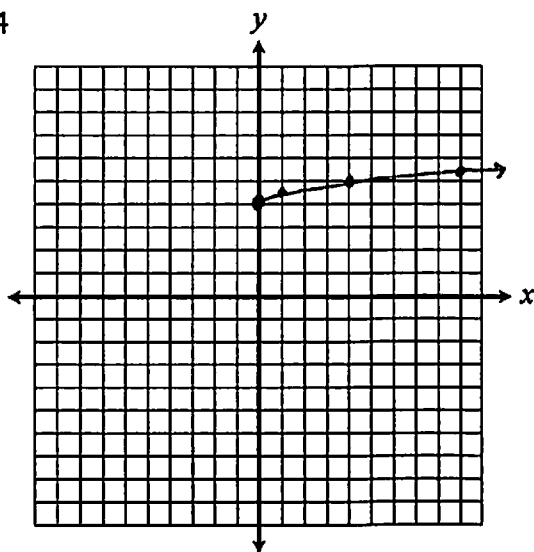
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Inc. Intervals: $(-\infty, 2), (4, \infty)$

Dec. Intervals: $(2, 4)$

75. $f(x) = \frac{1}{2}\sqrt{x} + 4$



Domain: $\{x | x \geq 0\}$

Range: $\{y | y \geq 4\}$

Endpoint/Turning Point: $(0, 4)$

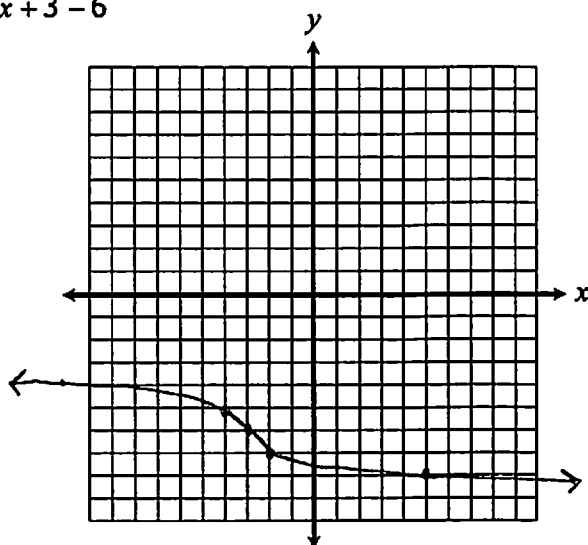
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow 0$, $f(x) \rightarrow 4$

Inc. Intervals: $(0, \infty)$

Dec. Intervals: —

76. $f(x) = -\sqrt[3]{x+3} - 6$



Domain: $\mathbb{R}$

Range: $\mathbb{R}$

Endpoint/Turning Point: $(-3, -6)$

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

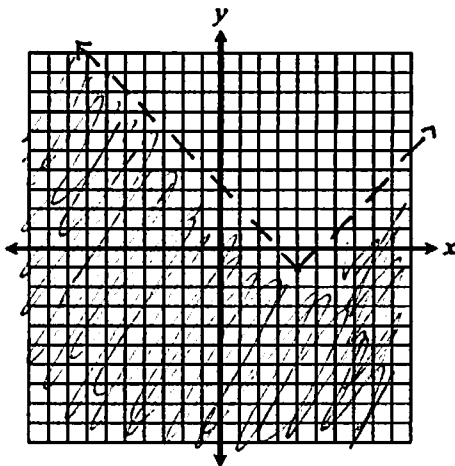
Inc. Intervals: —

Dec. Intervals: $(-\infty, \infty)$

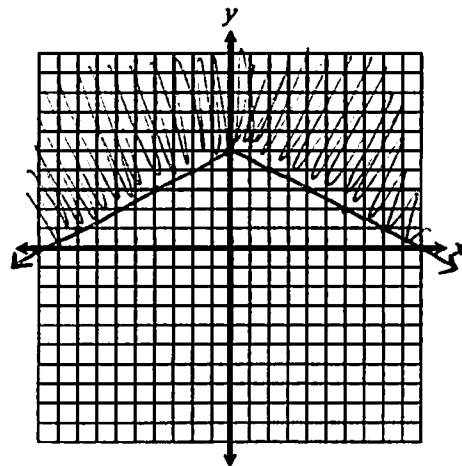
Topic 12: Graphing Absolute Value Inequalities

Graph each inequality.

77. $f(x) < |x - 4| - 1$



78. $f(x) \geq -\frac{1}{2}|x| + 5$



Topic 13: Applications & Regression

79. This past Saturday, Jack worked for 3 hours mowing the lawn and doing chores in the house. Mowing the lawn burns 352 calories per hour and doing chores burns 136 calories per hour. If he burned a total of 570 calories, how long did it take him to mow the lawn?

let x = mow

let $(3-x)$ = chores

$$352x + 136(3-x) = 570$$

$$352x + 408 - 136x = 570$$

$$216x = 162$$

$$x = 3/4$$

Mowing would take 45 min.

80. The pilot in a helicopter tossed a life float down to a swimmer in the water. The height, h , of the float after time t seconds is represented by the equation $h(t) = -16t^2 - 3t + 55$. Find the time it takes the float to reach the swimmer. $h=0$

$$-16x^2 - 3x + 55 = 0$$

$$x = \frac{3 \pm \sqrt{(-3)^2 - 4(-16)(55)}}{2(-16)}$$

$$x = \frac{3 \pm \sqrt{9 + 3520}}{-32}$$

$$x = \frac{3 \pm 59.405}{-32}$$

It would take
1.76 seconds.

-1.95
- (1.76)

81. The table below shows the number of teachers, t , and the number of students, s , at 6 high schools in the city. Write a **linear** equation to model the data, then predict the number of teachers for a school with 1,500 students.

t	115	140	76	168	110	92
s	1,122	1,390	788	1,564	1,081	924

$$f(x) = 8.68x + 130.36$$

158 teachers

$$1500 = 8.68x + 130.36$$

$$1369.64 = 8.68x$$

$$157.79 = x$$

82. The table below shows the population, p , of a city during certain years, t . Write a **cubic** equation to model the data, then predict the population of the city in 2020. $x = 30$

	3	6	10	15	19	24
t	1993	1996	2000	2005	2009	2014
s	2,850	3,257	4,234	6,475	10,158	27,385

$$f(x) = 6.86x^3 - 186.25x^2 + 1697.95x - 1070.43$$

$$f(30) = 67463$$

Topic 14: Functions

Use $f(x) = x^2 + 5x - 14$, $g(x) = x + 1$, and $h(x) = 15 - 2x^2$ to answer questions 83 – 88.

83. Find $[f(7)] - 2$

$$\begin{aligned} f(7) &= (7)^2 + 5(7) - 14 \\ &= 49 + 35 - 14 \\ &= 84 - 14 \\ &= 70 \end{aligned}$$

$$70 - 2 = \boxed{68}$$

84. If $g(x) = -23$, find x .

$$x + 1 = -23$$

$$\boxed{x = -24}$$

85. Find $(f - g)(x)$

$$(x^2 + 5x - 14) - (x + 1)$$

$$= \boxed{x^2 + 4x - 15}$$

86. Find $(f \cdot h)(x)$

$$(x^2 + 5x - 14) \cdot (15 - 2x^2)$$

$$= 15x^2 - 2x^4 + 75x - 10x^3 - 210 + 28x^2$$

$$= \boxed{-2x^4 - 10x^3 + 43x^2 + 75x - 210}$$

87. Find $\left(\frac{g}{f}\right)(x)$

$$\frac{x+1}{x^2+5x-14}$$

$$= \frac{x+1}{(x+7)(x-2)} ; x \neq -7, 2$$

88. Find $(h \circ g)(x)$

$$15 - 2(x+1)^2$$

$$= 15 - 2(x^2 + 2x + 1)$$

$$= 15 - 2x^2 - 4x - 2$$

$$= \boxed{-2x^2 - 4x + 13}$$

Use $f(x) = x^2 + 5x - 14$, $g(x) = x + 1$, and $h(x) = 15 - 2x^2$ to answer questions 89 – 90.

89. Find $(f+h)(-2)$

$$f(-2) = (-2)^2 + 5(-2) - 14$$

$$= 4 - 10 - 14 = \underline{-20}$$

$$h(-2) = 15 - 2(-2)^2$$

$$= 15 - 8 = \underline{7}$$

$$-20 + 7 = \underline{-13}$$

90. Find $g(f(8))$

$$f(8) = (8)^2 + 5(8) - 14$$

$$= 64 + 40 - 14$$

$$= 90$$

$$g(90) = 90 + 1 = \boxed{91}$$

Find the inverse of each function. Graph to show the relationship.

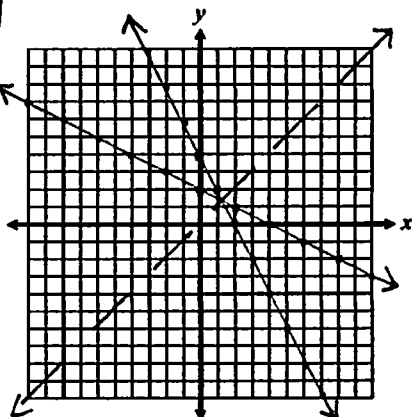
91. $f(x) = 4 - 2x$

$$f^{-1}(x) = -\frac{1}{2}x + 2$$

$$x = 4 - 2y$$

$$x - 4 = -2y$$

$$-\frac{1}{2}x + 2 = y$$



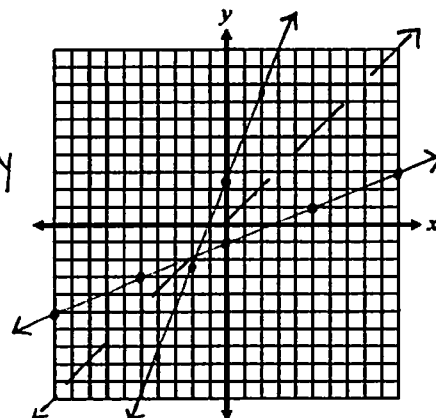
92. $f(x) = \frac{2}{5}x - 1$

$$f^{-1}(x) = \frac{5}{2}x + \frac{5}{2}$$

$$x = \frac{2}{5}y - 1$$

$$x + 1 = \frac{2}{5}y$$

$$\frac{5}{2}x + \frac{5}{2} = y$$



Determine whether the pair of functions are inverse functions. Prove algebraically.

93. $f(x) = -6x - 14$ and $g(x) = -\frac{1}{6}x - \frac{7}{3}$

$$f(g(x)) = -6\left(-\frac{1}{6}x - \frac{7}{3}\right) - 14$$

$$= x + 14 - 14$$

$$= x \quad \checkmark$$

yes, inverses

$$g(f(x)) = -\frac{1}{6}(-6x - 14) - \frac{7}{3}$$

$$= x + \frac{7}{3} - \frac{7}{3}$$

$$= x \quad \checkmark$$

94. $f(x) = \sqrt[3]{x+8}$ and $g(x) = (x-8)^3$

$$f(g(x)) = \sqrt[3]{(x-8)^3 + 8}$$

$$= \sqrt[3]{(x-8)(x^2 - 16x + 64) + 8}$$

$$= \sqrt[3]{x^3 - 24x^2 + 192x - 512 + 8}$$

Not
inverses!

$$= \sqrt[3]{x^3 - 24x^2 + 192x - 504}$$

Name: Key! Date: _____ Per: _____

Algebra 2: Mid-Year Test

- | | | | | |
|--------------|--------------|--------------|--------------|--------------|
| 1. <u>D</u> | 13. <u>D</u> | 25. <u>D</u> | 37. <u>A</u> | 49. <u>C</u> |
| 2. <u>A</u> | 14. <u>A</u> | 26. <u>C</u> | 38. <u>C</u> | 50. <u>B</u> |
| 3. <u>A</u> | 15. <u>B</u> | 27. <u>A</u> | 39. <u>D</u> | 51. <u>A</u> |
| 4. <u>B</u> | 16. <u>C</u> | 28. <u>D</u> | 40. <u>C</u> | 52. <u>B</u> |
| 5. <u>C</u> | 17. <u>B</u> | 29. <u>B</u> | 41. <u>A</u> | 53. <u>D</u> |
| 6. <u>D</u> | 18. <u>D</u> | 30. <u>C</u> | 42. <u>D</u> | 54. <u>D</u> |
| 7. <u>A</u> | 19. <u>B</u> | 31. <u>A</u> | 43. <u>C</u> | 55. <u>C</u> |
| 8. <u>B</u> | 20. <u>D</u> | 32. <u>C</u> | 44. <u>B</u> | 56. <u>D</u> |
| 9. <u>C</u> | 21. <u>C</u> | 33. <u>C</u> | 45. <u>D</u> | 57. <u>C</u> |
| 10. <u>C</u> | 22. <u>B</u> | 34. <u>B</u> | 46. <u>A</u> | 58. <u>B</u> |
| 11. <u>B</u> | 23. <u>A</u> | 35. <u>A</u> | 47. <u>D</u> | 59. <u>B</u> |
| 12. <u>C</u> | 24. <u>C</u> | 36. <u>B</u> | 48. <u>A</u> | 60. <u>A</u> |

Check All Answers!

Name: _____

Algebra 2

Date: _____ Bell: _____

Mid-Year Test

SHOW ALL WORK NEEDED TO ANSWER EACH QUESTION! Good Luck! ☺

1. Simplify completely: $\sqrt[3]{320x^{12}y^5}$ $\sqrt[3]{64x^{12}y^3} \quad \sqrt[3]{5y^2}$ $4x^4y \sqrt[3]{5y^2}$ A. $8x^9y \sqrt[3]{5y^2}$ B. $8x^4y \sqrt[3]{5y^2}$ C. $4x^9y^2 \sqrt[3]{5y^2}$ D. $4x^4y \sqrt[3]{5y^2}$	2. Simplify completely: $\sqrt{98m^5} + 4\sqrt{18m^5}$ $\sqrt{49m^4} \sqrt{2m} + 4\sqrt{9m^4} \sqrt{2m}$ $7m^2\sqrt{2m} + 12m^2\sqrt{2m}$ A. $19m^2\sqrt{2m}$ B. $19m^4\sqrt{2m}$ C. $8m^2\sqrt{29m}$ D. $8m^5\sqrt{29}$
3. Simplify completely: $\sqrt[4]{14k^3} \cdot \sqrt[4]{8k^5}$ $\sqrt[4]{112k^8}$ $\sqrt[4]{16k^8} \quad \sqrt[4]{7}$ $2k^2 \sqrt[4]{7}$ A. $2k^2 \sqrt[4]{7}$ B. $2k^4 \sqrt[4]{7}$ C. $2k^3 \sqrt[4]{7k^3}$ D. $4k^6 \sqrt[4]{7k^3}$	4. Simplify completely: $(2 + \sqrt{10})(7 + \sqrt{10})$ $14 + 2\sqrt{10} + 7\sqrt{10} + \sqrt{100}$ $14 + 9\sqrt{10} + 10$ A. $14 + 2\sqrt{10}$ B. $24 + 9\sqrt{10}$ C. $14 + 9\sqrt{10}$ D. $114 + 9\sqrt{10}$
5. Simplify completely: $\frac{\sqrt[3]{-270p^{14}}}{\sqrt[3]{2p^2}}$ $\sqrt[3]{-135p^{12}}$ $\sqrt[3]{-27p^{12}} \sqrt[3]{5}$ $-3p^4 \sqrt[3]{5}$ A. $-3p^4 \sqrt[3]{10}$ B. $-3p^3 \sqrt[3]{10p}$ C. $-3p^4 \sqrt[3]{5}$ D. $-3p^3 \sqrt[3]{5p}$	6. Simplify completely: $\frac{\sqrt{2x}}{\sqrt{27}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$ $\frac{\sqrt{6x}}{\sqrt{81}}$ A. $\frac{\sqrt{2x}}{3}$ B. $\frac{\sqrt{2x}}{9}$ C. $\frac{\sqrt{6x}}{27}$ D. $\frac{\sqrt{6x}}{9}$

7. Simplify completely: $\frac{6(5+\sqrt{7})}{(5-\sqrt{7})(5+\sqrt{7})}$ $\frac{30 + 6\sqrt{7}}{25 + 5\sqrt{7} - 5\sqrt{7} - \sqrt{49}}$ $\frac{30 + 6\sqrt{7}}{18}$ ☒ A. $\frac{5+\sqrt{7}}{3}$ ☐ B. $\frac{5+\sqrt{7}}{6}$ ☐ C. $\frac{5+\sqrt{7}}{18}$ ☐ D. $15+3\sqrt{7}$	8. Which expression is equivalent to $\sqrt[3]{8x^{12}y^5}$? $2x^{12/3}y^{5/3}$ ☐ A. $2x^4y^3$ ☒ B. $2x^4y^{5/3}$ ☐ C. $\frac{3}{8}x^4y^3$ ☐ D. $\frac{3}{8}x^4y^5$
9. Which expression is equivalent to $\left(m^{\frac{16}{3}}\right)^{\frac{1}{4}}$? $m^{\frac{16}{12}}$ $m^{4/3}$ $\sqrt[3]{m^4}$ ☐ A. $\sqrt[4]{m^3}$ ☐ B. $m^4\sqrt{m^3}$ ☒ C. $m^3\sqrt{m}$ ☐ D. $m^2\sqrt[3]{m}$	10. Which expression is equivalent to $\frac{9^{\frac{5}{2}}}{9^4} 2^{8/2}$? $9^{-3/2} = \frac{1}{\sqrt{9^3}}$ $= \frac{1}{3^3}$ ☐ A. 27 ☐ B. 3 ☒ C. $\frac{1}{27}$ ☐ D. $\frac{1}{3}$
11. Which expression is equivalent to $4\sqrt{-36}$? $4 \cdot 6i$ ☐ A. $72i$ ☒ B. $24i$ ☐ C. $12i$ ☐ D. $144i$	12. Which expression is equivalent to $-13i^{75}$? $-13(i^4)^{18}(i^3)$ $-13(1)(-i)$ ☐ A. 13 ☐ B. -13 ☒ C. $13i$ ☐ D. $-13i$
13. Simplify the expression below: $(7-2i)^2 - (25-12i)$ $(7-2i)(7-2i) - (25-12i)$ $49 - 28i + 4i^2 - 25 + 12i$ -4 $20 - 16i$ ☐ A. $28+21i$ ☐ B. $28-16i$ ☐ C. $20-40i$ ☒ D. $20-16i$	14. Simplify completely: $\frac{5(2+i)}{(2-i)(2+i)}$ $\frac{10+5i}{4+2i-2i-i^2} = \frac{10+5i}{5}$ ☒ A. $2+i$ ☐ B. $\frac{10+5i}{3}$ ☐ C. $2-i$ ☐ D. $\frac{10-5i}{3}$

15. Which set of numbers does not contain the value below? $(i\sqrt{3})^2 \quad i^2 \cdot 9$ -9 A. Complex Numbers ☒ B. Pure Imaginary Numbers C. Rational Numbers D. Integers	16. Which statement is false? A. All real numbers are complex numbers. B. All irrational numbers are real numbers. ☒ C. All complex numbers are pure imaginary numbers. D. All whole numbers are rational numbers.
17. Which statement illustrates the associative property of addition? A. $(1 + \sqrt{3})^2 + \sqrt{7} = 1 + (\sqrt{3} + \sqrt{7})^2$ ☒ B. $(-2 + 7i) + 4i = -2 + (7i + 4i)$ C. $\sqrt{18} + 5\sqrt{3} = \sqrt{3}(\sqrt{6} + 5)$ D. $(5i \cdot 2i) + 11i = 11i + (5i \cdot 2i)$	18. Identify the multiplicative inverse of the value below. $\left(-\frac{2i}{9}\right)$ A. $\frac{2i}{9}$ C. $-\frac{9i}{2}$ B. $\frac{9}{2i}$ ☒ D. $-\frac{9}{2i}$
19. Which is the complete factorization of the expression $81w^8 - 36w^4$? $9w^4(9w^4 - 4)$ $9w^4(3w^2 + 2)(3w^2 - 2)$ A. $9w^4(9w^4 - 4)$ ☒ B. $9w^4(3w^2 + 2)(3w^2 - 2)$ C. $(9w^4 + 6w^2)(9w^4 - 6w^2)$ D. $9w^4(9w + 1)(w - 4)$	20. Which of the following is the factored form of $x^3 + 343$? $(x + 7)(x^2 - 7x + 49)$ A. $(x + 7)^3$ B. $(x + 7)(x^2 + 49)$ C. $(x + 7)(x^2 + 7x + 49)$ ☒ D. $(x + 7)(x^2 - 7x + 49)$
21. For which polynomial is the expression $(2y - 3)$ a factor? A. $12y - 8$ $4(3y - 2)$ B. $45y^2 - 20$ $5(9y^2 - 4) = 5(3y - 2)(3y + 2)$ ☒ C. $16y^3 - 54$ $2(8y^3 - 27) = 2(2y - 3)(4y^2 + 6y + 9)$ D. $2y^2 + y - 3$ $(2y + 3)(y - 1)$	22. Which of the following expressions is not a factor of $12m^5 - 9m^3 - 3m$? $3m(4m^4 - 3m^2 - 1)$ $3m(4m^2 + 1)(m^2 - 1)$ $3m(4m^2 + 1)(m + 1)(m - 1)$ A. $3m$ ☒ B. $(m^2 + 1)$ C. $(4m^2 + 1)$ D. $(m - 1)$

23. Find the quotient: $(x^3 - 9x^2 + 5x - 4) \div (x - 2)$

$$\begin{array}{r|rrrr} 2 & 1 & -9 & 5 & -4 \\ & \downarrow & 2 & -14 & -18 \\ \hline & 1 & -7 & -9 & -22 \end{array}$$

☒ A. $x^2 - 7x - 9 - \frac{22}{x-2}$

B. $x^2 - 11x + 27 - \frac{58}{x-2}$

C. $x^2 - 7x + 19 - \frac{42}{x-2}$

D. $x^2 - 11x - 17 - \frac{38}{x-2}$

24. Solve: $8|3x - 5| = 56$

$$|3x - 5| = 7$$

$$3x - 5 = 7$$

$$3x = 12$$

$$x = 4 \checkmark$$

$$3x - 5 = -7$$

$$3x = -2$$

$$x = -\frac{2}{3} \checkmark$$

A. $\{-4\}$

B. $\{-4, \frac{2}{3}\}$

☒ C. $\{-\frac{2}{3}, 4\}$

D. $\{4, -\frac{14}{3}\}$

25. What of the following represents the solution to the inequality below?

$$|x - 7| + 12 \leq 25$$

$$|x - 7| \leq 13$$

$$x - 7 \leq 13$$

$$x \leq 20$$

$$x - 7 \geq -13$$

$$\text{and } x \geq -6$$

A. $(-\infty, -20]$ or $[6, \infty)$

B. $(-\infty, -6]$ or $[20, \infty)$

C. $[-20, 6]$

☒ D. $[-6, 20]$

26. Which of the following describes the roots of the function $f(x) = -x^2 - 3x + 1$?

$$(-3)^2 - 4(-1)(1)$$

$$9 + 4$$

$$13$$

A. exactly 1 rational root

B. 2 distinct rational roots

☒ C. 2 distinct irrational roots

D. 2 distinct imaginary roots

27. Solve: $-2x^2 + 12 = 68$

$$-2x^2 = 56$$

$$x^2 = -28$$

$$x = \sqrt{-28}$$

$$x = \pm 2i\sqrt{7}$$

☒ A. $\{\pm 2i\sqrt{7}\}$

B. $\{\pm 2i\sqrt{10}\}$

C. $\{\pm 2\sqrt{7}\}$

D. $\{\pm 2\sqrt{10}\}$

28. Solve: $x^2 - 12x - 14 = 0$

$$x^2 - 12x + 36 = 14 + 36$$

$$(x-6)^2 = 50$$

$$x-6 = \pm 5\sqrt{2}$$

$$x = 6 \pm 5\sqrt{2}$$

A. $\{-6 \pm i\sqrt{22}\}$

B. $\{6 \pm i\sqrt{22}\}$

C. $\{-6 \pm 5\sqrt{2}\}$

D. $\{6 \pm 5\sqrt{2}\}$

29. Which equation has a solution set of

$$\left\{-\frac{3}{2}, 4\right\}?$$

$$2x^2 - 5x - 12 = 0$$

$$(2x+3)(x-4) = 0$$

A. $2x^2 = 12 - 5x$

B. $3x^2 - 12 = x^2 + 5x$

C. $3x^2 - 9x = x + 8$

D. $2x^2 + 10x + 3 = 11 - x^2$

$$x = -\frac{3}{2} \quad x = 4$$

30. Which is a solution to the equation $x^3 - 64 = 0$?

$$(x-4)(x^2 + 4x + 16) = 0$$

$$x = 4$$

$$x = \frac{-4 \pm \sqrt{(4)^2 - 4(1)(16)}}{2(1)}$$

$$x = \frac{-4 \pm \sqrt{16 - 64}}{2}$$

$$x = \frac{-4 \pm \sqrt{-48}}{2}$$

$$x = \frac{-4 \pm 4i\sqrt{3}}{2}$$

A. $x = -4$

B. $x = -8$

C. $x = -2 + 2i\sqrt{3}$

D. $x = -2 + 2\sqrt{3}$

31. Which is not a solution to the equation below?

$$x^3 + 7x^2 - 9x - 63 = 0$$

$$x^2(x+7) - 9(x+7) = 0$$

$$(x^2 - 9)(x+7) = 0$$

$$x = \pm 3 \quad x = -7$$

A. 9

B. -7

C. 3

D. -3

32. A polynomial function has a zero at $x = 4i$. Which of the following expressions must be a factor of the polynomial?

A. $(x^2 + 8)$

B. $(x^2 - 8)$

C. $(x^2 + 16)$

D. $(x^2 - 16)$

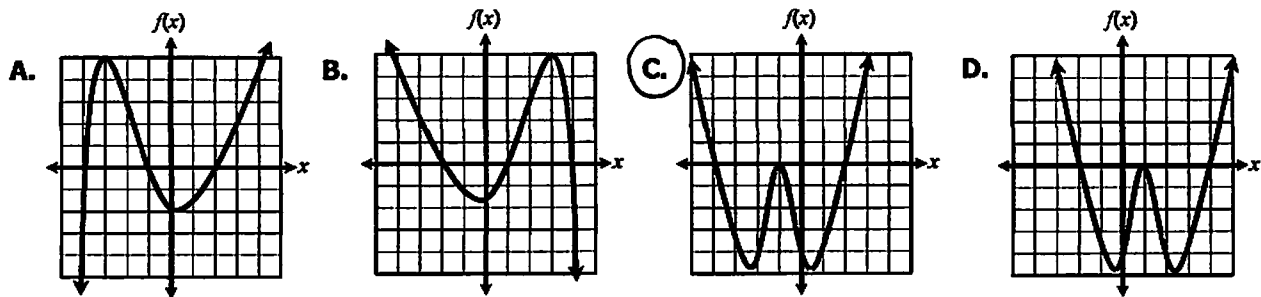
$$x^2 + 16 = 0$$

$$x^2 = -16$$

$$x = \pm 4i$$

intersect bounce intersect

33. A polynomial function has a zeros at -4 (multiplicity 3), -1 (multiplicity 8) and 2 (multiplicity 1). Which equation could represent this function?



34. Solve: $\sqrt{12-2x}+3=7$

$$\sqrt{12-2x} = 4$$

$$12-2x = 16$$

$$-2x = 4$$

$$x = -2$$

A. $x = -17$

B. $x = -2$

C. $x = -14$

D. $x = 16$

35. Solve: $\sqrt[4]{\frac{1}{3}x-8}=2$

$$\frac{1}{3}x - 8 = 16$$

$$\frac{1}{3}x = 24$$

$$x = 72$$

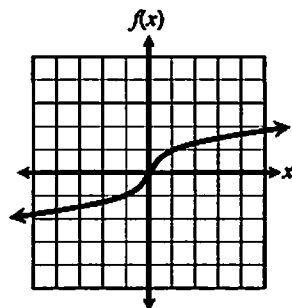
A. $x = 72$

B. $x = 8$

C. $x = \frac{10}{3}$

D. $x = -2$

36. The graph of a parent function is shown below. Which function belongs to the same family?



A. $f(x) = \sqrt{x} + 6$

B. $f(x) = \sqrt[3]{x-4}$

C. $f(x) = 2x^3 + 5$

D. $f(x) = (x-1)^2$

37. The graph of the absolute value parent function is shifted 3 units left, 8 units down, then vertically stretched. Which equation could represent this function?

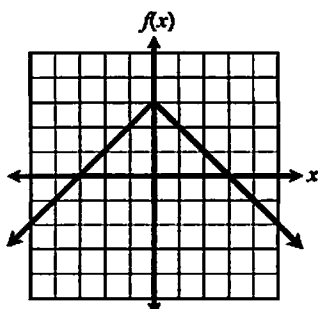
A. $f(x) = 2|x+3| - 8$

C. $f(x) = 2|x-3| - 8$

B. $f(x) = \frac{1}{2}|x+3| - 8$

D. $f(x) = \frac{1}{2}|x-3| - 8$

38. Which equation defines the function graphed below?



A. $f(x) = -|x+3|$

B. $f(x) = -|x-3|$

C. $f(x) = -|x| + 3$

D. $f(x) = -|x| - 3$

39. Which function has two distinct real zeros?

A. $f(x) = 2x + 2$

B. $f(x) = (x-2)^2$

C. $f(x) = x^3 + 2$

D. $f(x) = |x| - 2$

40. Which of the following describes the range of the function below?

$$f(x) = -(x+3)^2 - 5$$

- A. $\{y \mid y \leq 3\}$
- B. $\{y \mid y \geq 3\}$
- ☒ C. $\{y \mid y \leq -5\}$
- D. $\{y \mid y \geq -5\}$

41. Which function does not have a domain of all real numbers?

- ☒ A. $f(x) = \sqrt{x} - 1$
- B. $f(x) = -x^3 + 4x$
- C. $f(x) = -|x-5| + 1$
- D. $f(x) = \frac{1}{2}\sqrt[3]{x} + 6$

42. Identify the vertex of the function below.

$$f(x) - 4 = (x+1)^2$$

$$f(x) = (x+1)^2 + 4$$

- A. $(-4, 1)$
- B. $(1, -4)$
- C. $(-1, -4)$
- ☒ D. $(-1, 4)$

43. Which of the following statements is true regarding the function below?

$$f(x) = (x-3)(x-7)^2$$

- A. $(3, 0)$ is a zero and a relative minimum
- B. $(3, 0)$ is a zero and a relative maximum
- ☒ C. $(7, 0)$ is a zero and a relative minimum
- D. $(7, 0)$ is a zero and a relative maximum

44. The graph of the cubic parent function is reflected across the x -axis. Which of the following describes the end behavior of the new function as x approaches negative infinity?

- A. $f(x)$ approaches negative infinity
- ☒ B. $f(x)$ approaches positive infinity
- C. $f(x)$ approaches 0
- D. $f(x)$ approaches 3

45. Which function family can have more than one increasing interval?

- A. Square Root
- B. Cube Root
- C. Quadratic
- ☒ D. Cubic



46. During which interval is the function below only decreasing?

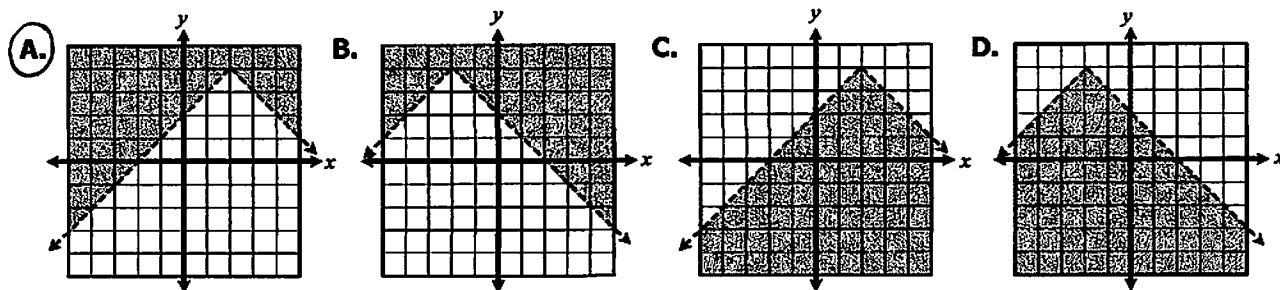
$$f(x) = (x+2)(x-1)^2$$

- ☒ A. $-1 < x < 1$
- B. $-2 < x < 1$
- C. $-1 < x < 2$
- D. $-\infty < x < 1$

47. The graph of a quadratic function increases through the interval $-\infty < x < 5$, then decreases through the interval $5 < x < \infty$. If the vertex of the graph is located at $(5, 9)$, which equation could represent this function?

- A. $f(x) = (x+5)^2 + 9$
 B. $f(x) = (x-5)^2 + 9$
 C. $f(x) = -(x+5)^2 + 9$
 D. $f(x) = -(x-5)^2 + 9$

48. Which graph represents the inequality $f(x) > -|x-2| + 4$?



49. Kate's soccer team is selling coupon books and recipe books to raise money for transportation to their games. Coupon books cost \$12 each and recipe books cost \$7.50 each. If Kate sold a total of 42 books and raised \$441, how many coupon books did she sell?

$$\begin{aligned} x + y &= 42 \rightarrow y = 42 - x \\ 12x + 7.5y &= 441 \\ \hline 12x + 7.5(42 - x) &= 441 \\ 12x + 315 - 7.5x &= 441 \\ x &= 28 \end{aligned}$$

- A. 18 coupon books
 B. 24 coupon books
 C. 28 coupon books
 D. 14 coupon books

50. A baseball is thrown from a height of 5 feet. The height, h , of the ball at time t seconds is modeled by the equation $h(t) = -16t^2 + 100t + 5$. How long will it take the ball to reach the ground?

$$\begin{aligned} -16t^2 + 100t + 5 &= 0 \\ t &= 6.3 \end{aligned}$$

- A. 5.8 seconds
 B. 6.3 seconds
 C. 7.2 seconds
 D. 7.6 seconds

51. The table below shows the yearly average selling price of a home in the Orchard Woods subdivision during certain years. Using a quartic model, approximate the selling price of a home in Orchard Woods in 2020. $x = 22$

	Year	Average Selling Price
0	1998	\$144,000
5	2003	\$174,600
12	2010	\$175,120
14	2012	\$180,500
17	2015	\$203,000

$$f(x) = -0.05x^4 + 58.2x^3 - 1482.5x^2 + 12083x + 144000$$

$$f(22) = \$300,303.4$$

- (A) \$300,303
B. \$306,128
C. \$310,502
D. \$315,457

Use the functions $f(x) = 4x + 1$, $g(x) = 2x^2 - 6x$, and $h(x) = -x^2 + 3x - 19$ for questions 52-57.

52. Find $-2 \cdot [g(-2)]$

$$g(-2) = 2(-2)^2 - 6(-2) \\ = 20$$

- A. 8
(B) -40
C. 20
D. -16

53. If $f(x) = -13$, find x .

$$4x - 1 = -13 \\ x = -\frac{7}{2}$$

- A. -3
B. $-\frac{2}{7}$
C. -51
(D) $-\frac{7}{2}$

54. Find $(h - g)(x)$

$$(-x^2 + 3x - 19) - (2x^2 - 6x) \\ -x^2 + 3x - 19 - 2x^2 + 6x$$

- A. $x^2 - 9x - 19$
B. $3x^2 - 9x + 19$
C. $-3x^2 - 3x - 19$
(D) $-3x^2 + 9x - 19$

55. Find $(f \cdot h)(x)$

$$(4x + 1)(-x^2 + 3x - 19) \\ -4x^3 + 12x^2 - x^2 + 3x - 76x - 19$$

- A. $-x^2 + 12x + 3x - 18$
B. $-4x^3 + 6x^2 - 11x - 18$
(C) $-4x^3 + 11x^2 - 73x - 19$
D. $-4x^3 + 12x^2 - 76x - 19$

56. Which describes the domain of the function

$$\left(\frac{g}{f}\right)(x)? \quad \frac{2x^2 - 6x}{4x + 1}$$

- A. All Real Numbers
B. $\left\{x \mid x \neq \frac{1}{4}\right\}$
C. $\{x \mid x \neq -1\}$
(D) $\left\{x \mid x \neq -\frac{1}{4}\right\}$

57. Find $(g \circ f)(5)$

$$f(5) = 21 \\ g(21) = 756$$

- A. 81
B. 420
(C) 756
D. 915

58. Find the inverse of the function below.

$$f(x) = 8x - 4$$

$$x = 8y - 4$$

$$\frac{1}{8}x + \frac{1}{2} = y$$

A. $f^{-1}(x) = 2x + 1$

C. $f^{-1}(x) = 2x - 1$

☒ B. $f^{-1}(x) = \frac{1}{8}x + \frac{1}{2}$

D. $f^{-1}(x) = \frac{1}{8}x - \frac{1}{2}$

59. The point $(-7, 1)$ lies on the graph of $g(x)$. Which point must lie on the graph of $g^{-1}(x)$?

A. $(7, -1)$

☒ B. $(1, -7)$

C. $(-1, 7)$

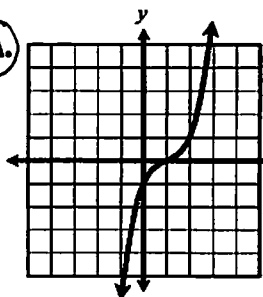
D. $(-7, -1)$

60. Given $f(x) = \sqrt[3]{x} + 1$, which graphs represents $f^{-1}(x)$?

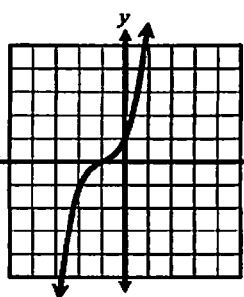
$$x = \sqrt[3]{y} + 1$$

$$(x-1)^3 = y$$

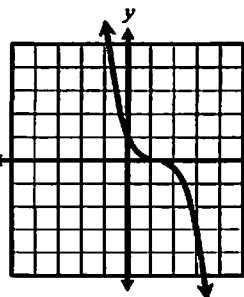
☒ A.



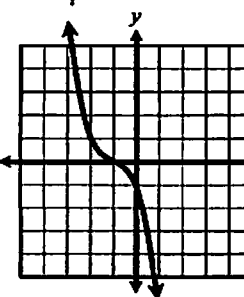
B.



C.



D.



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